China Graduate Innovation and Practice Competition Series

"Huawei Cup" 21st China Graduate Mathematical Modeling Competition

Title: Research on Spatio-Temporal Analysis Mathematical Models of Geographic Big Data Based on Multiple Intelligent Algorithms

Abstract

China is subject to the combined influence of multiple Earth-system-sphere factors—including topography, atmospheric circulation, land-sea thermal contrasts, biological processes, and human activities—forming an intricate human-land coupled system. With advances in science and technology, it has become increasingly feasible to construct geographic system models using geographic big data combined with algorithms, enabling an integrated understanding of geographic phenomena. Our team aims to approach the problem through an Earth-systems thinking framework, applying machine learning algorithms to build a model-data coupling system to solve the following problems:

For Problem 1, this paper describes the spatio-temporal evolution characteristics of precipitation and land use types. The annual intra-annual **standard deviation** of daily precipitation and annual **mean** precipitation are used to assess precipitation variability; a **stacked chart** is used to visualize **seasonal changes**; and five-year-interval **national precipitation distribution maps** for 1990–2020 are produced, with **200, 400, and 800 mm isohyetal lines** drawn to analyze spatial distribution patterns. For land use types, the **single land-use dynamic degree** and **comprehensive land-use dynamic degree** are used to measure rates of change, and a **land-use transfer matrix** provides an intuitive view of the degree of change.

For Problem 2, a **Gumbel distribution** (an asymmetric extreme-value distribution widely used in geography for natural hazard analysis) is first fitted to precipitation data from the past decade to define **extreme weather thresholds** dominated by heavy rainfall, thereby deriving the annual frequency of extreme weather events at each grid point. The role of terrain-climate interaction in extreme weather formation is then explored; a **binary linear regression** model establishes a terrain-climate interaction model. Due to collinearity among elevation, precipitation, and temperature, multiple **machine learning algorithms** (such as **Random Forest** and **Support Vector Regression**) are applied for model construction and compared for stability and accuracy. The Random Forest model performs best. A **feature importance chart** is exported to deeply explore **the influence of terrain-climate interaction on extreme weather formation**.

For Problem 3, rainfall, terrain, and land use are first quantified as independent variables to build a **logistic regression model** for analyzing the **threshold conditions for storm-induced disasters**.

To quantify the effects of different land use types on storm resistance and disaster probability, a **Kostiakov infiltration model** based on hydrology and soil science builds a disaster index evaluation system for five land use types, yielding land-use coefficients. To forecast storm-disaster-vulnerable areas from 2025 to 2035, three methods are used: Method 1 applies **ARIMA time-series forecasting** for precipitation and linear regression for land-use coefficients; Method 2 uses a **Random Forest prediction model**; Method 3 is based on an **LSTM neural network**. Predicted values are substituted into the logistic regression model, using **storm disaster probability** to indicate **capacity to cope with storm disasters**. Results for 2025–2035 are visualized on maps, **marking the most vulnerable areas**.

For Problem 4, the influences of **physical-geographic parameters** (e.g., temperature, elevation, precipitation) and **human-geographic parameters** (e.g., GDP, population) on land use type transitions are explored. The discussion area is divided into four parts **using the Qinling-Huaihe Line and the "Hu Huanyong Line" as boundaries**. A model describing "the characteristics and structure of China's land use change" is built with **multiple indicators**. Combined with the **Entropy Weight-TOPSIS evaluation method**, the factors influencing the per-capita arable land ratio coefficient in different regions are analyzed. Finally, **multiple linear regression** and a **Random Forest model** are applied, with **Spearman correlation** optimization analysis to reveal the relationship between regional characteristics and per-capita arable land.

Keywords: Random Forest; Gumbel Distribution; ARIMA Forecasting; Entropy Weight-TOPSIS Method; Mixed Linear Regression; Logistic Regression

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I. Problem Restatement

1.1 Problem Background

China is vast in territory and large in population, spanning five temperature zones from tropical to subarctic and encompassing seven climate types. Its coastline exceeds 32,600 km in total length; the enormous land-sea thermal contrast makes monsoons a critical factor influencing China's climate, population, and economy. The elevation range is approximately 9,003 m, encompassing plateaus, mountains, plains, basins, and hills. Combined with the rapid development of human activities and the socioeconomic environment, natural systems and the geographic systems of human activity are tightly coupled [1], with factors from all spheres interrelated and interacting, forming China's intricate human-land system. For a long time, from county/city to national scales, high-accuracy description, simulation, and prediction of individual factors (e.g., temperature, precipitation), multi-factor combinations, or coupled systems have been complex, costly, and difficult. However, in recent years, with multidisciplinary integration and advances in technology and computing power, it has become possible to use big data with machine learning and other algorithms, focusing on an integrated perspective to extract, integrate, deep-learn, and simulate factors from multiple spheres—including the atmosphere, hydrosphere, lithosphere, and anthroposphere—thereby constructing geographic system models from multiple angles and in a comprehensive manner.

Geographic system models are the most important research tools for understanding and predicting changes in the patterns and processes of geographic systems at different scales [1]. This paper aims to systematically describe the spatio-temporal distribution characteristics of precipitation and land use types across mainland China, extracting several influencing factors—precipitation, elevation, temperature, longitude, and latitude—from complex interactions among climate, terrain, and human activities, then normalizing them for machine learning. The Random Forest model, as one of the better choices among existing machine learning methods for handling massive high-dimensional data, can learn from historical data to summarize patterns of change and predict future trends under corresponding parameters. We therefore hope to use the models constructed in this study to establish an extreme weather forecasting and meteorological disaster early-warning system, providing references for climate change adaptation and disaster prevention and mitigation policies; at the same time, to establish a model-data fusion system for China's land use types through machine learning, providing scientific decision support for China's future sustainable and high-quality development.

1.2 Problem Statement

Problem 1:

Based on specified datasets, extract, process, analyze, and visualize the corresponding data. Use concise statistical indicators or charts to describe and summarize the spatio-temporal evolution characteristics of precipitation and land use/land cover types across mainland China from 1990 to 2020.

Problem 2:

Using terrain, rainfall, and temperature datasets, extract feature factors, build a mathematical model, and quantitatively characterize the role of terrain-climate interaction in the formation of extreme weather events.

Problem 3:

First, based on analysis of rainfall, terrain, and land use type data—in which rainfall has the strongest spatio-temporal variability and uncontrollability; land use, as the integrated result of natural conditions and human activities, also evolves spatio-temporally but is to some degree controllable; and terrain is the most stable and difficult-to-change factor—while accounting for differences in spatio-temporal variability, determine the threshold conditions for storm-induced disasters in combination with the mathematical model established in Problem 2. Then, combining the historical spatio-temporal evolution characteristics of rainfall and land use/land cover changes from Problem 1, forecast the regions with the weakest capacity to cope with storm disasters in China from 2025 to 2035, and present the forecast results in map form.

Problem 4:

For the land use/land cover situation at the intersection of physical geography and human geography, incorporating its "characteristics" in the description, estimation, and prediction tasks of the first three problems, use geographic big data to build a corresponding mathematical model to simplify and synthesize the data, describing the characteristics and structure of China's land use change. Explain and validate your summary from both accuracy and utility perspectives.

II. Problem Analysis

Problem 1:

This problem requires describing and summarizing the spatio-temporal evolution characteristics of precipitation and land use/land cover types across mainland China from 1990 to 2020, based on specified datasets.

Precipitation, as a highly continuous variable, requires multi-scale temporal description covering intra-annual, inter-annual, and decadal scales. To accurately describe intra-annual variability and inter-annual oscillations, the annual intra-annual standard deviation of daily precipitation and the annual mean precipitation are introduced for assessment. Due to the influence of monsoons on China's climate, precipitation varies significantly with seasons; stacked charts are therefore used to

visualize seasonal precipitation trends. Decadal variation characteristics are captured through changes in annual mean precipitation. For the spatial distribution description, five-year-interval national precipitation distribution heat maps are exported using ArcMap, combined with 200 mm, 400 mm, and 800 mm isohyetal lines, to describe the spatial variation trends of China's precipitation over 30 years.

Land use/land cover types, as variables with abrupt changes and discrete distributions, do not exhibit as pronounced short-term fluctuations as precipitation. The single land-use dynamic degree and comprehensive land-use dynamic degree are therefore introduced as statistical indicators, focusing on the rates and magnitudes of change for different land use types at five-year intervals, to describe the spatio-temporal evolution of land use/land cover types.

Problem 2:

This problem requires building a mathematical model based on specified terrain, temperature, and precipitation datasets to explain the role of terrain-climate interaction in extreme weather formation. We first use a binary linear regression model to explore correlations between terrain factors and climate factors, establishing a terrain-climate interaction model. We then select appropriate terrain-climate coupling factors, while also using the Gumbel distribution to define and delimit extreme weather events. Multiple models—including Random Forest, SVR, and multiple linear regression—are built for learning and fitting, and their stability and accuracy are compared to provide a reference for model selection in subsequent problems.

Problem 3:

This problem requires determining the threshold conditions for storm-induced disasters by considering the roles of terrain, rainfall, and land use type in the transition from "heavy rainfall" to "disaster" as reflected in Problem 2, along with their interactions. It also requires forecasting the regions most vulnerable to storm disasters in China from 2025 to 2035, combining the historical spatio-temporal evolution characteristics of rainfall and land use/land cover changes from Problem 1. Analysis of the characteristics of these three variables reveals that rainfall has the strongest spatio-temporal variability and uncontrollability; land use, as the integrated result of natural conditions and human activities, is somewhat controllable as it evolves spatio-temporally; while terrain is the most stable and difficult-to-change factor, and its rate of temporal change can be approximately ignored. Based on this understanding, we first screen small-scale factors characterizing these three geographic elements based on knowledge from meteorology, hydrology, and soil science. After disaster determination, a logistic regression model is used to establish a disaster index evaluation system to quantify regional vulnerability. Simultaneously, based on the spatio-temporal variation rules of geographic elements from Problem 1 and the optimal model selection from Problem 2, a Random Forest model is used to forecast changes in precipitation and land use over the next decade. The disaster index evaluation system is then applied for prediction, and results are visualized on maps.

Problem 4:

This problem requires building a corresponding mathematical model using geographic big data to simplify and synthesize the data, describing the characteristics and structure of China's land use change, and validating the summary from both accuracy and utility perspectives. To this end, we intend to use the constructed models—combined with physical-geographic parameters (e.g., temperature, elevation, precipitation) and human-geographic parameters (e.g., GDP, population)

—to explore the transitions of land use types and their characteristics. The discussion area is divided into four sub-regions using the Qinling-Huaihe Line and the "Hu Huanyong Line" as boundaries. A model describing "the characteristics and structure of China's land use change" is established, and its accuracy and applicability in the human-land coupled system are explored.

III. Model Assumptions and Notation

3.1 Basic Model Assumptions

(1) All natural and human-geographic variables (temperature, land use types, GDP, population, etc.) in the datasets are assumed to be accurate measured values.
(2) During the forecast period, it is assumed there are no drastic disturbances caused by human factors such as large-scale warfare or major policy changes.
(3) During the forecast period, it is assumed that no severe tectonic movements (e.g., large-scale earthquakes) cause significant changes to China's topography.
(4) The training sample size is assumed to be sufficiently large and the time series sufficiently long to reflect classification criteria and long-term trend changes.

3.2 Notation

Table 3-1 Notation

Symbol	Meaning	Unit
P	Precipitation	mm
T	Temperature	℃
E	Elevation	m
t	Time	min
$I(t)$	Infiltration rate	mm/min
S	Area	m^2
η	Comprehensive land-use dynamic degree	min^{-1}
K	Single land-use dynamic degree	min^{-1}
GDP	Gross Domestic Product	10,000 CNY/m^2
pop	Population	persons
MSE	Mean Squared Error	—
R^2	Coefficient of determination	—

IV. Model Development and Solution

4.1 Problem 1: Descriptive System Construction

4.1.1 Problem 1 — Description of Spatio-Temporal Evolution of Precipitation

Based on the uniformity of raster cells, we use Python to load and extract the daily precipitation per unit area per unit raster cell (spatial resolution: 0.25°×0.25°, with 36,864 cells in total) from the "Daily Gridded Precipitation Dataset for Mainland China" [2], denoted Pd. Based on China's markedly uneven seasonal climate, daily precipitation data are processed into quarterly mean precipitation and annual mean precipitation using the following formula:

$$a_i = P_d \cdot S_g$$

$$\alpha = \sum_{i=1}^{n} a_i$$

$$A = \sum \alpha$$

$$S = \sum S_g$$

$$P = \frac{A}{S} = \frac{\Sigma \alpha}{S} = \frac{\sum_{j=1}^{m} \sum_{i=1}^{n} P_d \cdot S_g}{\sum_{j=1}^{m} S_g}$$

where a_i is the daily mean precipitation of a unit raster cell; α is the annual mean precipitation of a unit raster cell; n is the total number of days in the current quarter or year (Q2: n=91; Q3: n=92; Q4: n=92; standard year Q1: n=90, annual: n=365; leap year Q1: n=91, annual: n=366); P represents the annual mean precipitation of mainland China; A represents total national precipitation; S represents the land area of mainland China; m represents the total number of raster cells covering mainland China.

Generally, because the Earth is a slightly oblate irregular pear-shaped sphere, mathematical map projections produce some error and distortion, causing differences in raster cell area at the same spatial resolution across different orientations. However, since the effect on data calculation accuracy in this problem is small (e.g., at northern latitude vs. northern latitude xx), it is ignored when calculating annual mean precipitation, and the formula can be approximately simplified accordingly.

$$P = \sum_{j=1}^{m} \sum_{i=1}^{n} P_d$$

To quantify the cumulative amount and contribution of precipitation in each quarter of the year, we visualize quarterly mean precipitation using a stacked area chart, yielding the following statistical chart:

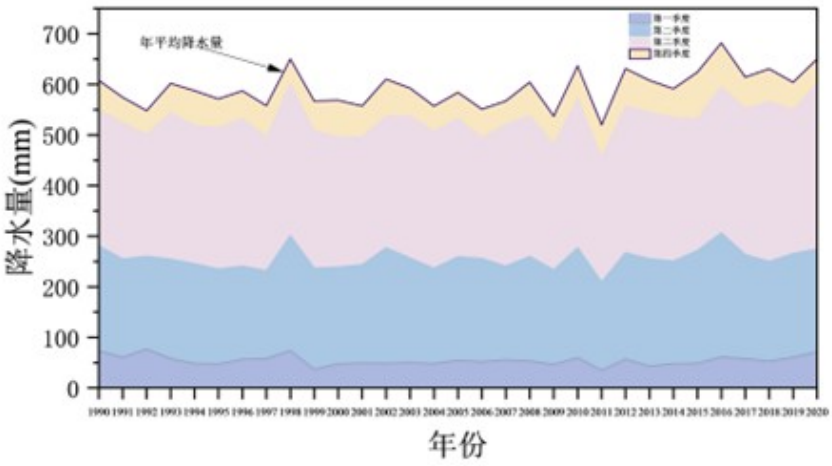


Figure 4-1 Multi-scale precipitation stacked statistical chart

Observations show that precipitation in mainland China exhibits clear and stable seasonal regularity, typically with concurrent rain and heat, showing a summer- and autumn-concentrated trend. The second and third quarters consistently have much higher precipitation than the first and fourth quarters, with Q3 slightly higher than Q2, highly consistent with China's characteristic of having the world's most typical monsoon climate. Meanwhile, although annual precipitation surged in 1998 and 2010 and was relatively low in 2009 and 2011, its decadal trend has generally been rising year by year, particularly from 2012 to 2020, reflecting an overall tendency of China's climate toward greater humidity.

To describe the intensity of intra-annual precipitation change and inter-annual variability, and to observe the magnitude of climate fluctuations over short and long time scales, we introduce the annual standard deviation of daily precipitation, calculated as follows:

$$\sigma p(t) = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}\left(P_i(t) - P_{avg}(t)\right)^2}$$

where $\sigma_p(t)$ represents the relative standard deviation of daily mean precipitation within a given year; i represents the total number of days in that year (365 for standard years, 366 for leap years); $P_{av}g(t)$ represents the mean of daily precipitation within that year; $P_i(t)$ represents the precipitation on a given day. By extracting year-by-year deviation values of annual mean precipitation from the multi-year average, combined with the annual mean precipitation and the relative standard deviation of intra-annual daily mean precipitation, the following statistical chart is obtained:

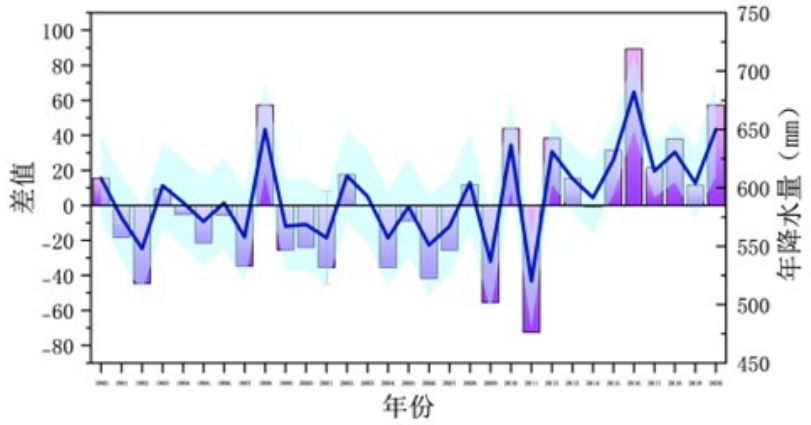


Figure 4-2 Annual mean precipitation composite statistical chart

In the figure, the purple bar chart represents the deviation of the annual mean precipitation from the multi-year average; the blue line represents the annual mean precipitation; the cyan area represents the relative standard deviation of daily mean precipitation for the corresponding year. As shown, due to China's climate characterized by distinct seasons and concentrated precipitation, the relative standard deviation has consistently been high over the years (14.79–19.65), with slight fluctuations during 1994–2009. Since 2010, this value has shown a continuous upward trend, coinciding with the period of continuously increasing global extreme weather frequency, indicating that the influence of global climate variability on China's precipitation has been intensifying year by year.

Isohyetal lines are an important concept in geography, reflecting the combined influences of land-sea thermal contrasts, topography, ocean currents, atmospheric circulation, and other complex factors on precipitation. The 800 mm annual isohyetal line generally serves as the boundary between China's humid and semi-humid regions and between subtropical and temperate monsoon climates; the 400 mm line commonly serves as the boundary between China's eastern monsoon region and the northwestern arid/semi-arid regions; and the 200 mm line is the boundary between arid and semi-arid regions, and between desert and non-desert areas. By observing their multi-year changes, a more geographically characteristic account of inter-annual climate variation trends in mainland China can be provided, and future evolutionary trends can be simulated and predicted.

We therefore introduce isohyetal lines combined with the national annual mean precipitation distribution for the same year, and use Python to visualize the spatio-temporal distribution evolution of China's precipitation from 1990 to 2020, yielding the following statistical charts:

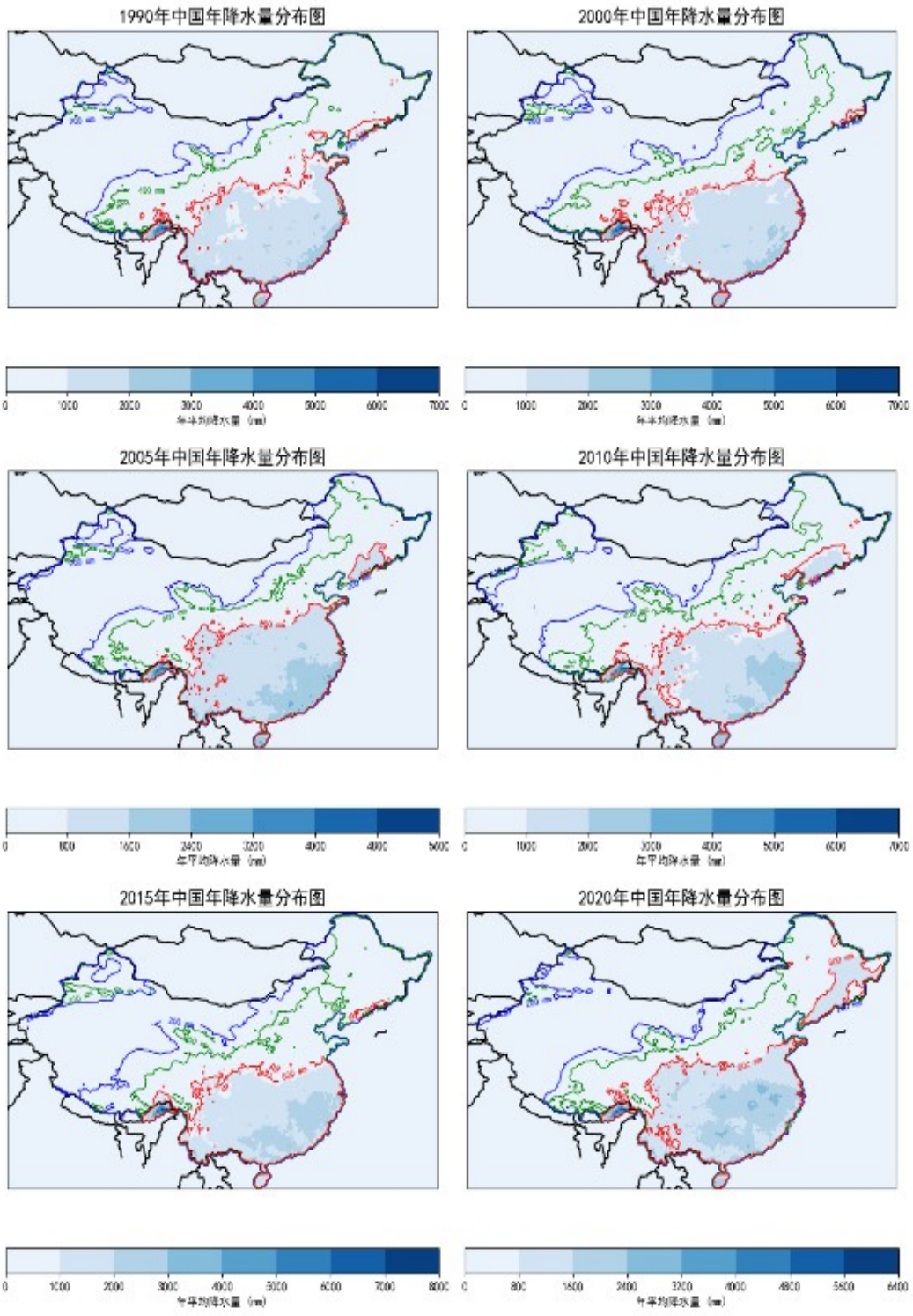


Figure 4-3 Annual mean precipitation–isohyetal line spatio-temporal variation chart

Note: The red line is the 800 mm isohyetal line; the green line is the 400 mm isohyetal line; the blue line is the 200 mm isohyetal line.

Observations of the above figure show that from 1990 to 2020, the spatial distribution of precipitation in China has been generally stable, with higher values in southeastern coastal areas and lower values in northwestern regions. Precipitation in China's southeastern coastal areas has been continuously increasing, and areas of abundant rainfall have been continuously expanding

northwestward, while the northwestern region has remained relatively dry over the years. From 1990 to 2015, all three isohyetal lines showed a southward-shifting trend, with a slight northward shift from 2015 to 2020. The movements also exhibit distinct segmented characteristics: all three lines show relative stability in the middle segment, slight fluctuation in the western segment, and the most pronounced fluctuation in the eastern segment.

4.1.2 Problem 1 — Description of Spatio-Temporal Variation Analysis of Land Use/Land Cover Types

Land use types are comprehensively influenced by terrain, climate, and human activities, exhibiting characteristic spatial distribution and spatio-temporal variation patterns. Using a 10-year change cycle from 1999 to 2009, the 1999, 2000, and 2019 land use/land cover type datasets are selected for data extraction, cleaning, and calculation. Python is used to visualize the spatial distribution patterns and evolution of five land use types, as shown below:

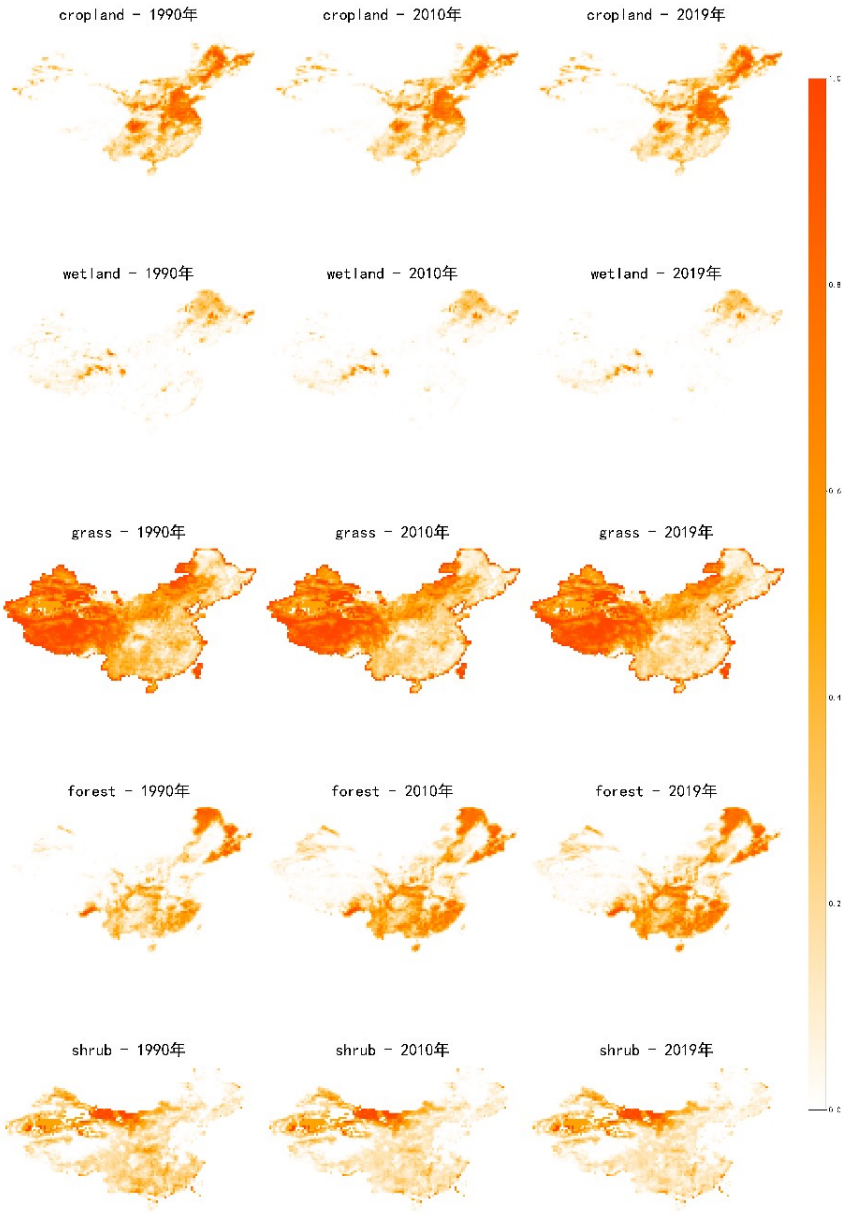


Figure 4-4 Coverage rates of five land use types in mainland China for 1999, 2000, and 2009

Observations reveal that China's arable land is mainly concentrated in the eastern region, where the terrain is relatively flat and the climate relatively suitable, favorable for farming and forming an important foundation of China's agricultural production. Forests are more widely distributed in the humid and semi-humid mountainous and hilly areas of the east, northeast, and southwest. Grassland and shrubland are widely distributed, mainly concentrated in the arid and semi-arid plateaus and mountainous areas of the west and northwest; grassland is more concentrated in the southwest, while shrubland is more concentrated in the north and northwest. Wetlands are mainly distributed in the northeastern regions and the Tibetan Plateau area.

As shown above, the spatial distribution of China's land use types is influenced by terrain and climate, exhibiting characteristic regional features. The spatio-temporal variation trends need to be described next. Land-use dynamic degree, as a statistic reflecting the rate and magnitude of change of different land use types within a study area over a given period, can quantitatively describe the magnitude and trend of China's land use type changes since 1990. The single land-use dynamic degree and comprehensive land-use dynamic degree are therefore introduced, with their respective calculation formulas as follows:

where K represents the single land-use dynamic degree for a given land use type; Wa and Wb represent the initial and final areas of that land use type within a given study period; and t is the duration of that study period.

η represents the comprehensive land-use dynamic degree; ΔW_{i-j} is the sum of areas of land use type i converted to other land use types within a given study period; W_i is the initial-period area of land use type i; and t is the duration of the study period.

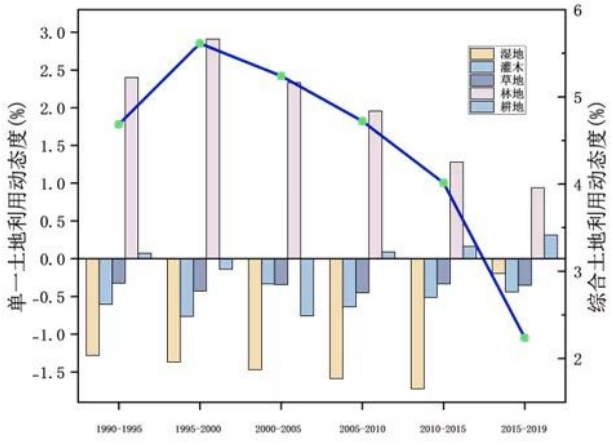


Figure 4-5 Land-use dynamic degree statistical chart

Observations reveal that over the past 30 years, the dynamic degrees of wetlands, shrubland, and grassland have all been on a declining trend. The rate of shrinkage of wetland area has been continuously accelerating, while the rates of shrinkage of shrubland and grassland have fluctuated. For arable land and forest, due to the implementation of policies returning farmland to forest, forest area has continuously increased from 1990 to 2020 with somewhat declining growth rates, while arable land area slowly declined from 1990 to 2005 and steadily increased from 2005 to 2020.

The land-use transfer matrix is a model based primarily on Geographic Information System (GIS) technology and spatial statistical analysis methods that describes the transition relationships among different land use types. It is commonly used to reveal the internal mechanisms of land use change and assess the impact of human activities on land use change, providing a scientific basis for related research and policy formulation. Its calculation principle quantitatively describes the transition relationships and amounts among various land use types within a given area over a certain time period. The following provides a detailed explanation:

Table 4-6 Land-Use Transfer Matrix Calculation Table

T_N / T_L	C_1	C_2	…	Cn	T_{LS}	T_{LO}
C_1	S_{11}	S_{12}	…	S_{1n}	S_{1*}	S_{*1}-S_{11}
C_2	S_{21}	S_{22}	…	S_{2n}	S_{2*}	S_{*2}-S_{22}
…	…	…	…	…	…	…
C_n	S_{n1}	S_{n2}	…	S_{nn}	S_{n*}	S_{n*}-S_{nn}
T_{NS}	S_{*1}	S_{*2}	…	S_{*n}		
T_{NI}	S_{*1}-S_{11}	S_{*2}- S_{22}	…	S_{*n}-S_{nn}		

where TL denotes the previous time period and TN denotes the next time period. C_1 through C_n represent n different land cover types. S_{ij} represents the area converted from type C_i (previous period) to C_j (next period); S_i* is the total area of type C_i in the previous period; $S*_j$ is the total area of type C_j in the next period. S_i*-S_{ii} is the outflow from type C_i. $S*_j-S_{jj}$ is the inflow to type C_j.

Here we use Python to calculate the spatial values, obtain the land-use transfer matrix, and visualize it using a Sankey diagram.

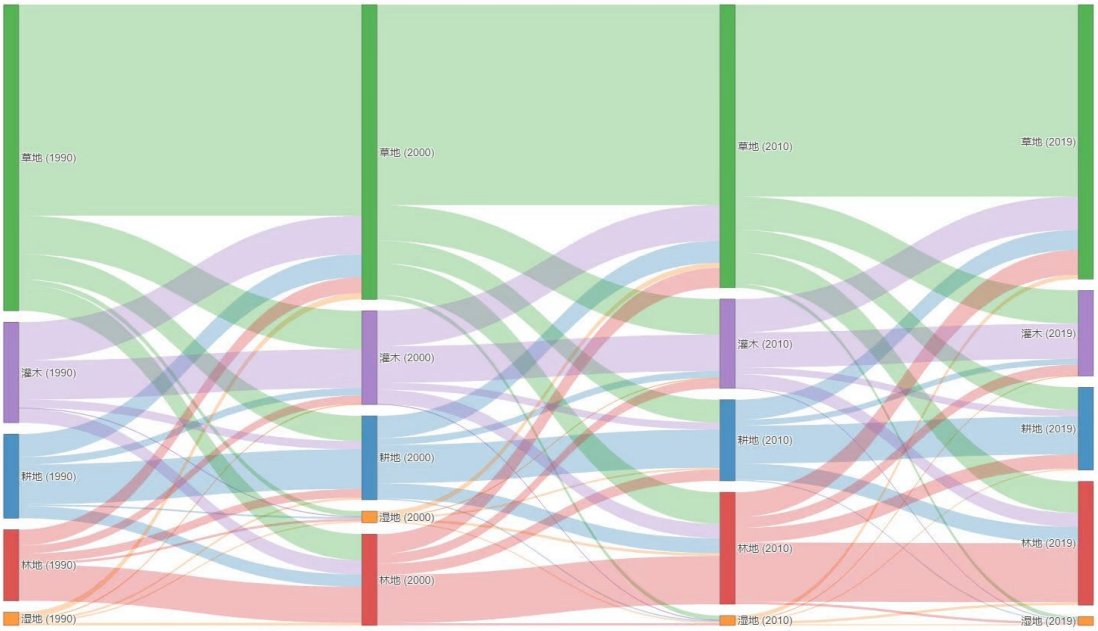


Figure 4-7 Land-Use Transfer Matrix (1990–2019)

4.2 Problem 2: Model Development and Solution

4.2.1 Problem 2 Solution Framework

(1) Data Processing: Since the precipitation, temperature, and terrain datasets [2-11] do not share a uniform spatial resolution, data cleaning is first performed to ensure correct CRS (Coordinate Reference System) information, remove missing values and invalid measurements (e.g., raster cells where all terrain factors equal zero), and normalize the data (Note: all data used subsequently come from the above datasets). The 2010 data cleaning results are shown below as an example:

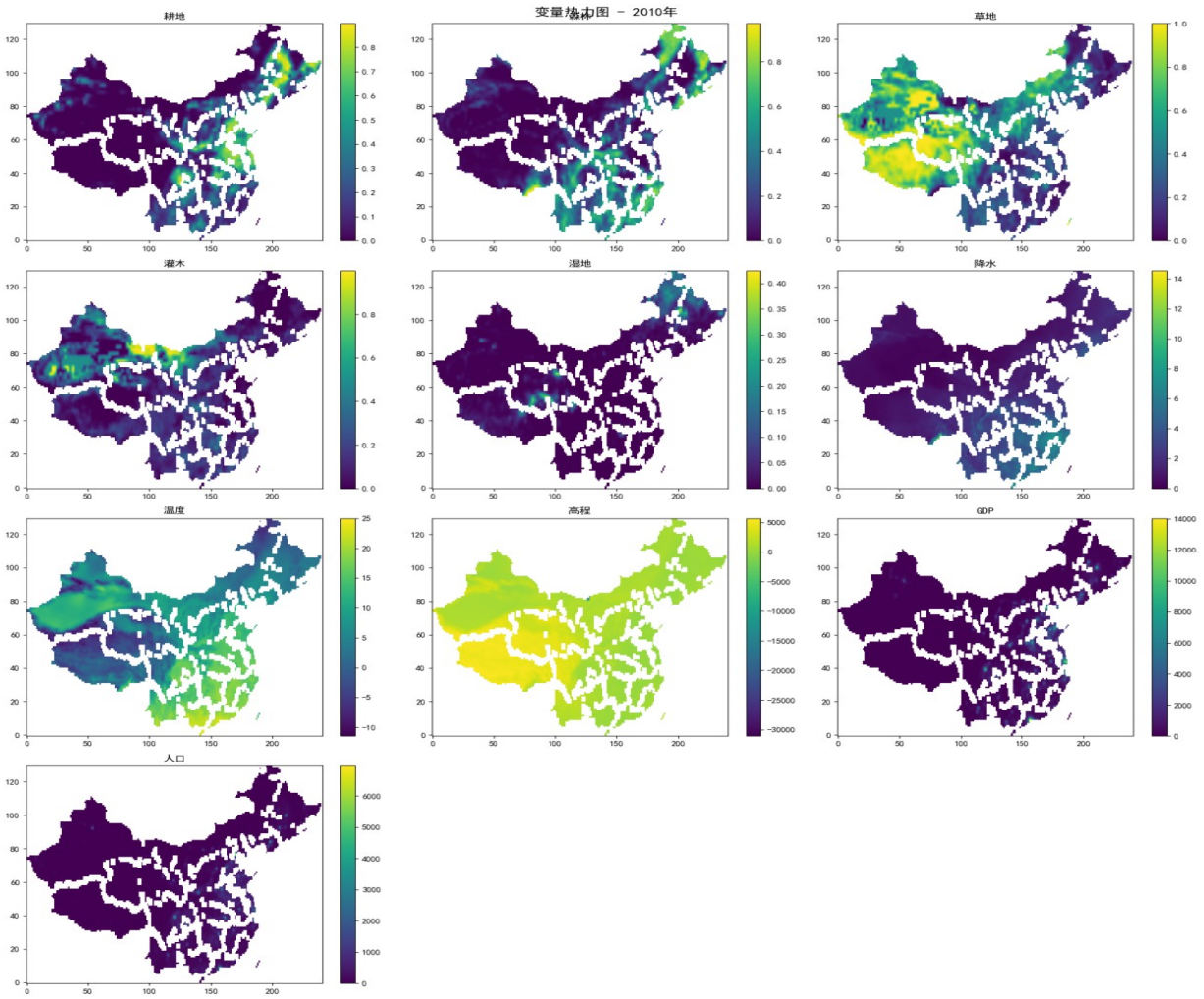


Figure 4-8 Multi-dataset cleaning illustration

For missing and anomalous values, resampling is applied to multiple datasets (precipitation, temperature, terrain, etc.) to achieve spatial alignment at a uniform resolution. Normalized precipitation, temperature, and terrain data are then extracted from TIFF files using Python.

(2) Conceptual Definition: This problem requires a mathematical definition of extreme weather events represented by heavy rainfall. In the evaluation framework for extreme climate events in geography, methods include the absolute threshold method, percentile method, and historical ranking method. This problem integrates the spatio-temporal distribution variation characteristics of terrain-climate elements and uses the percentile method to define extreme weather events: a fixed percentile value (e.g., the 80th or 95th percentile) from a long time series is selected as the threshold, and events exceeding this threshold are defined as extreme climate events. Annual maximum daily precipitation observations at each unit grid point—possessing the property of being independently and identically distributed—are extracted to form a sample set for fitting the Gumbel distribution. Maximum Likelihood Estimation (MLE) is used to estimate the Gumbel distribution parameters from annual maximum precipitation data, and the 95th percentile defines extreme precipitation events.

(3) Model Construction and Analysis: A binary linear regression model is used to fit the relationships among elevation, temperature, and precipitation to explore terrain-climate interactions. Multiple models (Random Forest, SVR, and multiple linear regression) are then constructed to build mathematical models of the relationship between terrain-climate coupling factors and extreme precipitation events, quantifying the role of terrain-climate interactions in extreme weather formation.

4.2.2 Problem 2 Model Construction, Solution, and Analysis

(1) Construction and Analysis of the Extreme Weather Classification Model

The Gumbel distribution, as one type of Generalized Extreme Value (GEV) distribution, is commonly used to model the distribution of maxima of independent and identically distributed random variables, and is widely applied in meteorology and environmental science to estimate the probability of extreme weather events (such as extreme precipitation, wind speed, and high temperatures). In this problem, we use the Gumbel distribution to build an extreme climate classification model represented by heavy rainfall. The PDF (probability density function) of the Gumbel distribution can be expressed as:

where β is the scale parameter (controlling the width of the distribution) and μ is the location parameter (closely related to the data mean, controlling the center of the distribution). The log-likelihood function of the Gumbel distribution can be expressed as:

where n is the total number of observations and x_i is the i-th observation. MLE is used in Python to optimize the model, fitting parameters from 1960–2015 annual maximum precipitation data, and a bisection method calculates the 95th percentile to define extreme precipitation events. The quantile calculation formula is:

where xq is the q-th quantile of this dataset. The fitted distribution results for this problem are shown below:

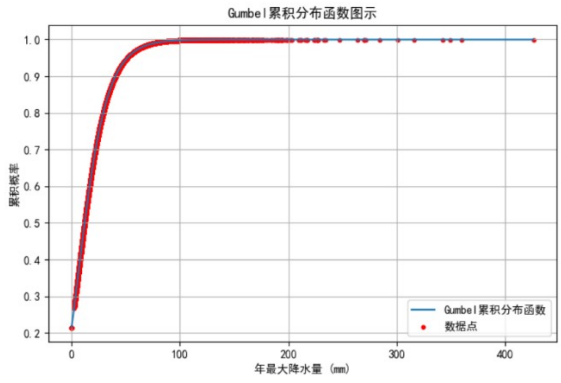


Figure 4-9 Gumbel distribution function illustration

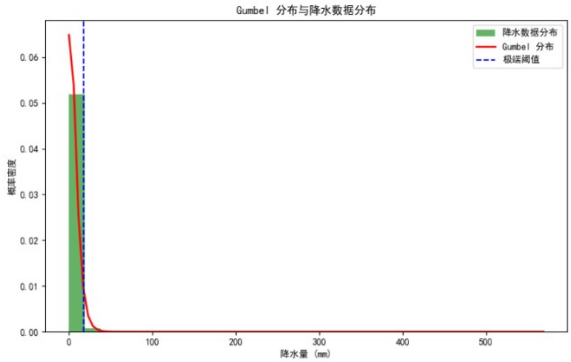


Figure 4-10 Gumbel function precipitation distribution fitting chart

After obtaining the extreme precipitation threshold, we use this threshold to identify and count precipitation events exceeding the threshold at each unit grid point and mark them as extreme precipitation events, counting the total annual frequency of extreme precipitation events at each unit grid point. To ensure geographic information consistency with other datasets, resampling achieves spatial alignment, and the frequency of extreme precipitation events along with corresponding grid spatial information is output as a CSV file to simplify subsequent data analysis.

(2) Construction and Analysis of the Multiple Linear Regression Model

Temperature and precipitation are the key elements describing climate, while elevation is an important factor characterizing the spatial variability of terrain. We first use a binary linear regression model to explore terrain-climate interactions. This problem assumes that precipitation P is a linear function of temperature T and elevation E:

where β_0 is the intercept term; β_1 and β_2 are the coefficients for temperature and elevation respectively; and ε is the error term. Parameters β_1 and β_2 are fitted using Ordinary Least Squares (OLS) (where y denotes precipitation, x_1 denotes temperature, x_2 denotes elevation):

Using Python to process and acquire the three datasets, the fitted result is:

$$P=5.967 - 0.259\ T - 0.001E$$

The scatter plot prediction simulation surface for this data is shown below:

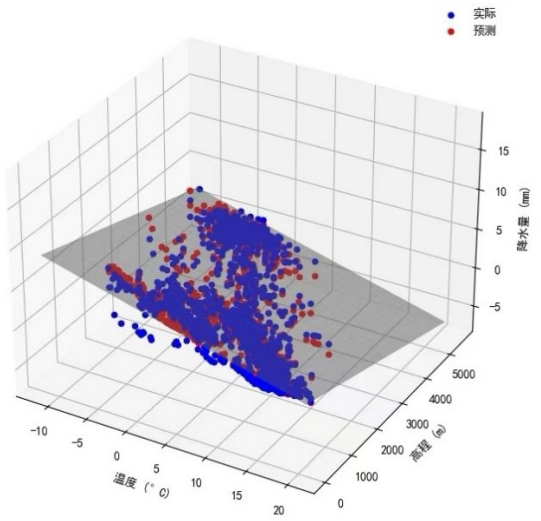


Figure 4-11 Binary linear regression scatter plot prediction simulation

Model evaluation is performed by checking whether the residuals satisfy assumptions such as normal distribution and independence. Residual statistical charts and Q-Q plots are drawn as follows:

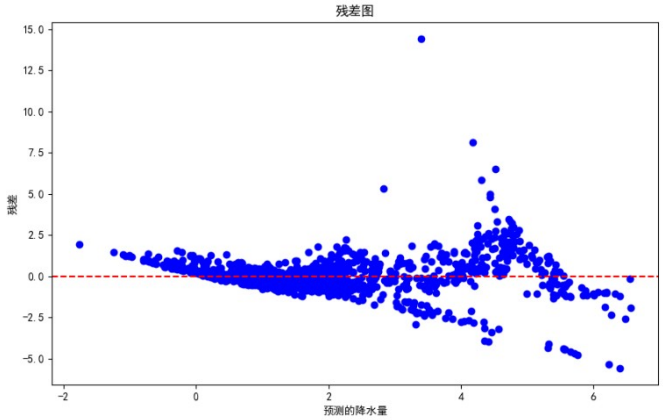


Figure 4-12 Binary linear regression scatter residual distribution chart

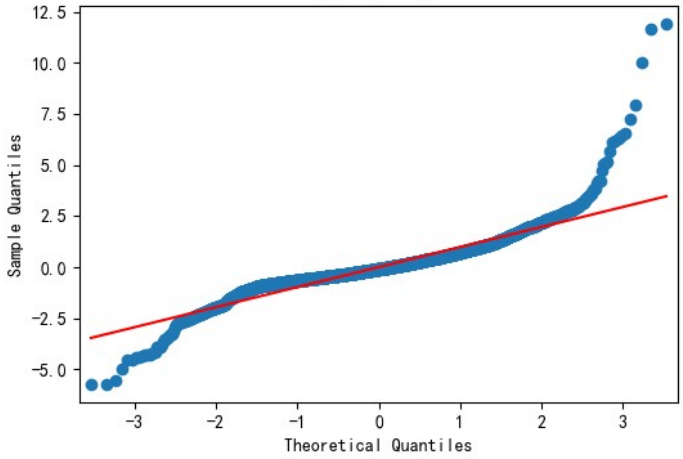


Figure 4-13 Binary linear regression residual Q-Q plot

As shown, most residual values are randomly distributed near the zero line and show no obvious pattern or trend, indicating that this model's fitting has no specific systematic error and is relatively uniform. Observing the residual Q-Q plot, most residual values fall on the line k=1, indicating that their distribution is close to normal; the model fits well and can relatively accurately explain terrain-climate interactions.

(3) Multi-Model Construction and Accuracy Assessment

Academia recognizes that extreme precipitation events are primarily influenced by the combined effects of local hydrological conditions, climate, terrain, and other factors. The factors describing these three geographic elements are numerous, including daily temperature range, daily mean temperature, daily mean precipitation, and elevation. Considering the representativeness of each element and the extraction difficulty and calculation accuracy in the corresponding datasets, elevation, temperature, and precipitation are selected as the main influencing factors. Extreme weather events are defined based on the Gumbel distribution calculations described above. Multiple models are constructed on a long time scale, comprehensively considering the spatial variability of each factor, to explore the role of terrain-climate interaction in extreme weather formation and to compare the accuracy and precision of each model. The main models are as follows:

(a) Multiple Linear Regression (MLR) Model

The construction principle of the multiple linear regression model is:

Arranging the above m independent variables in columns to construct the matrix:

where m is the number of independent variables and n is the total sample size. The r-th sample can be represented as The optimal undetermined coefficients are calculated using Ordinary Least Squares:

Based on the above understanding, the multiple linear regression model constructed in this problem can be expressed as:

where Y represents the frequency of extreme precipitation events; T represents temperature; β_0 is the intercept term; (x,y,t) represents spatial and temporal coordinates; β_1, β_2, β_3 are the regression coefficients for the corresponding climate factors; and ε is the error term. Substituting the processed datasets yields:

$$Y = 1.2058 + 0.0224 + 0.0356 + 0.0356$$

The model fit yields $R^2 = 0.302$, indicating poor accuracy. The suspected reason is a collinearity problem. When independent variables are highly correlated with each other, the multiple linear regression model's estimates may be unstable or distorted. This model is not suitable for simulation in this problem.

(b) Random Forest (RF) Model

The Random Forest model is a highly versatile and accurate ensemble learning model in machine learning. It builds multiple decision trees to form a decision forest and comprehensively considers the prediction results of each decision tree to improve the model's output stability and accuracy. The model's working principle is illustrated below:

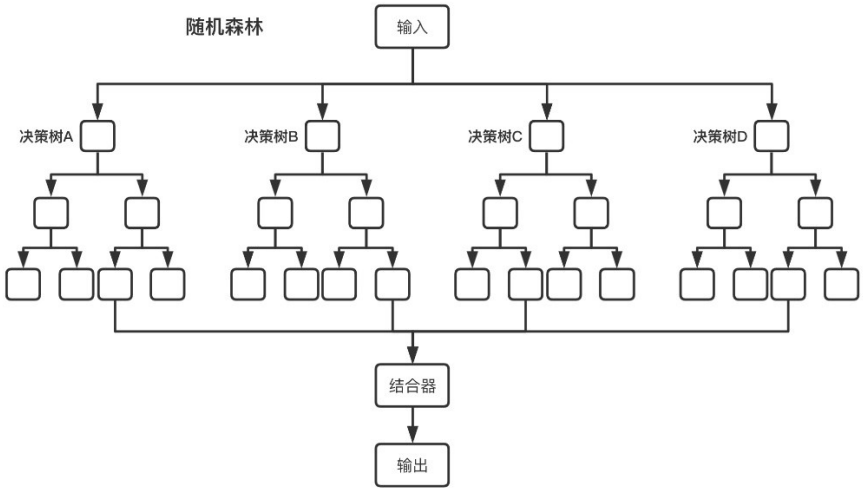


Figure 4-14 Random Forest model working principle illustration

First, this model performs random sampling with replacement from the input dataset to build multiple subsets, each serving as a training set for one decision tree, ensuring each tree may contain duplicate samples. The remaining data serve as the test set for evaluating accuracy and stability. At each node of each decision tree, a random subset of features is selected for optimal splitting to reduce correlations among models and improve overall model fit. Finally, the best feature is selected at each node until stopping conditions are met. In this problem, datasets of elevation, temperature, precipitation, and extreme precipitation events are extracted to build a Random Forest model. After fitting, the feature importance chart is output as follows:

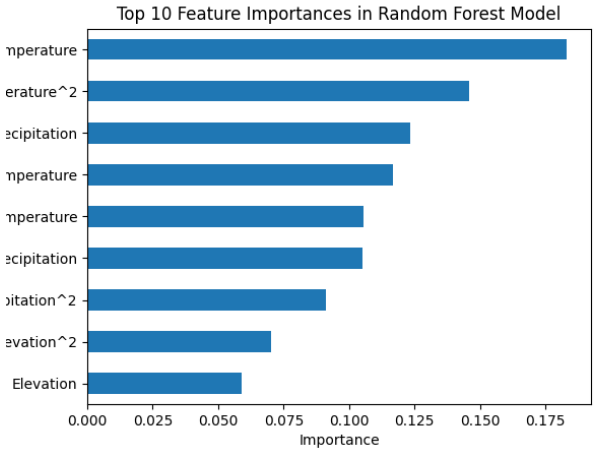


Figure 4-15 Random Forest feature importance chart

Feature importance can be estimated by calculating the average contribution of each feature to the reduction in impurity:

where p(t) is the proportion of samples reaching node t, and $\Delta i(s_t,t)$ is the reduction in impurity caused by the split s_t by feature j at node t.

For the model's goodness of fit, time-series cross-validation is used to evaluate performance. R^2 scores and MSE are calculated for each feature (land-use coefficient and precipitation). This model achieves R^2=0.88, indicating high fitting accuracy. The decision process of the first decision tree is shown below:

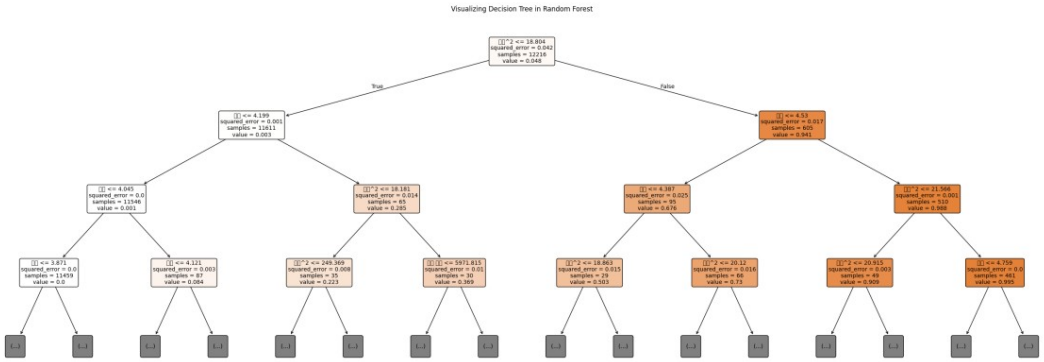


Figure 4-16 Random Forest model decision tree (first tree only shown)

Analysis of the Random Forest model fitting results: in the process of terrain-climate interaction in extreme weather formation, climate plays a relatively dominant role—temperature and precipitation contribute more to the formation of extreme weather. However, this does not mean elevation has no influence. The relatively few terrain factors extracted when building the model dataset may be a reason; during subsequent model optimization, more terrain factors (e.g., slope, aspect) should be extracted to further improve model fitting accuracy.

(b) Support Vector Regression (SVR) Model

The Support Vector Regression model, as a machine learning method based on statistical learning theory, is suitable for handling regression problems. This model finds a hyperplane to fit the data, adjusting parameters to minimize error around the hyperplane while maintaining a certain margin to enhance adaptability. Fitting yields R^2=0.09, indicating poor fitting accuracy. A comparison of the fitting accuracy of the three models is shown below:

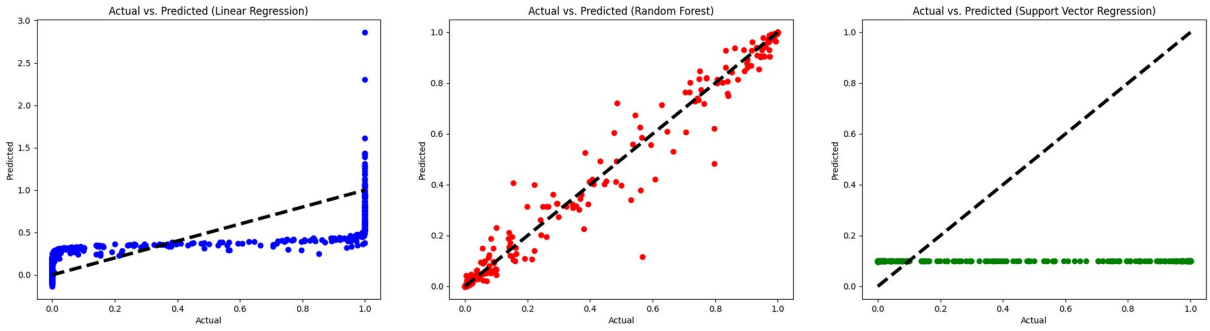


Figure 4-17 Comparison chart of fitting accuracy of three models

Note: Blue represents the linear regression model; red represents the Random Forest model; green represents the Support Vector Regression model.

Results show that the Random Forest model achieves the best fitting accuracy, while the SVR model and the multiple linear regression model perform poorly.

4.3 Problem 3: Model Development and Solution

4.3.1 Problem 3 Solution Framework

(1) Evaluation Model Construction

The threshold conditions for storm disaster formation are comprehensively influenced by multiple geographic factors including atmospheric circulation, terrain, and hydrological characteristics. This problem requires exploring the combined effects of rainfall, terrain, and land use on storm disaster formation and investigating the threshold conditions for disaster occurrence. Among these three factors: rainfall is strongly influenced by terrain and meteorological conditions, with strong spatio-temporal variability and uncontrollability; land use, as the integrated result of natural conditions and human activities, is somewhat controllable in its spatio-temporal evolutionary trend; while terrain is relatively the most stable and difficult to change.

Based on the model characteristics from Problem 2, we first select smaller-scale typical geographic factors to establish disaster gestation evaluation indicators for the three geographic elements of rainfall, terrain, and land use. At the same time, extreme precipitation disaster datasets are defined and extracted. A Logistic regression model is used for fitting to establish a "Storm Disaster Gestation Index" evaluation system (the higher the system value, the greater the probability of disaster formation).

(2) Forecasting Model Construction, Comparison, and Evaluation

Land-use coefficients and precipitation are planned to be forecasted over a long time series using a Long Short-Term Memory network (LSTM) model with the past 30 years of land use datasets. Due to the excessively large computation load and overly long processing time (>48h), this forecasting method had to be abandoned; the corresponding model code has been recorded in the Appendix. Some forecasting results are shown below—these are the 2025–2035 change trends for five randomly selected grid points:

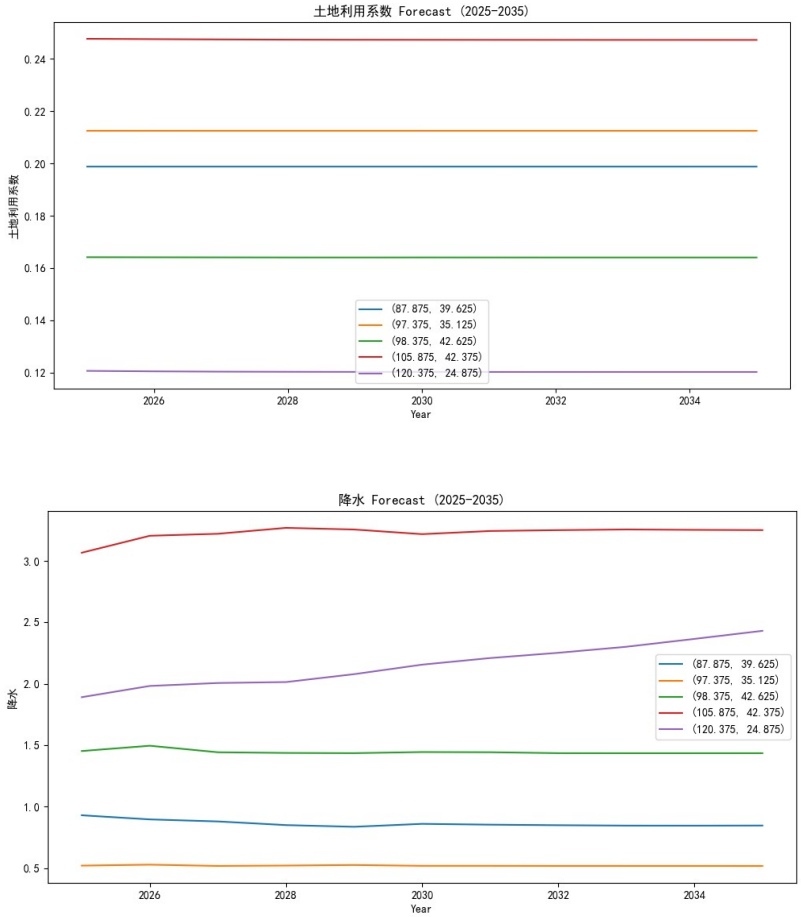


Figure 4-18 LSTM model partial fitting results display

Therefore, due to computational resource limitations, dimensionality reduction analysis is planned for unit grid point land use data. Based on the above forecasting data characteristics, random grid points are extracted from the dataset for linear regression analysis. Since the linear relationship between land-use coefficients and time is significant, linear regression is used for fitting and predicting the change trends of unit grid point land-use coefficients.

For the precipitation forecasting model construction, observing the descriptive characteristics from Problem 1 and the LSTM model fitting results, it is evident that historical spatio-temporal evolution of precipitation has periodic fluctuation characteristics. Therefore, the ARIMA model—better suited for handling random fluctuations or noise in time series—is adopted. Based on the autocorrelation and partial autocorrelation characteristics of the dataset, the optimal ARIMA model parameters (m, n, P) are fitted and determined, where m represents the order of the autoregressive term, n represents the number of differencing operations, and P represents the order of the moving average term. The forecasting results are shown below:

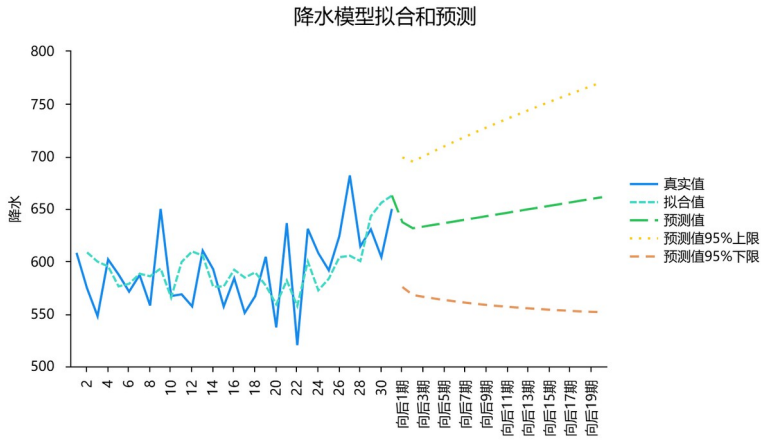
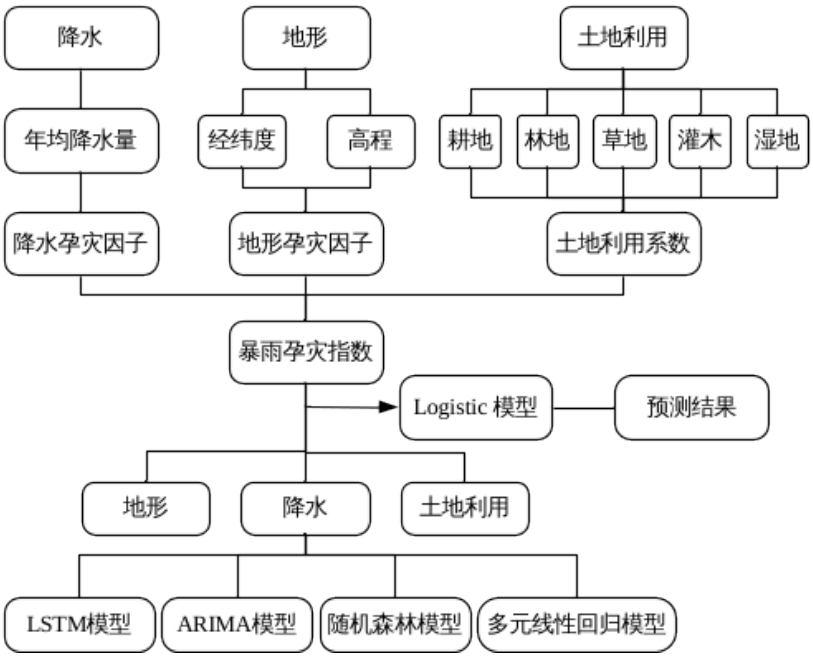


Figure 4-19 Spatio-temporal evolution forecast of precipitation based on the ARIMA model

Observations show that this model has high simulation accuracy. The ratio of its predicted annual mean values to the baseline values of this dataset (2015 is selected as the baseline year) is used to obtain the growth rate of annual mean precipitation at each unit grid point, constructing the forecasting model system.

Based on the model evaluation from Problem 2, the Random Forest model has high prediction accuracy. We therefore extract data from 144×256 unit grid points over the past 30 years for precipitation and land use, establish training and test sets, and use the Random Forest model to forecast precipitation factors and land-use coefficients.

The following is a mind map summarizing the solution approach for this problem:



4.3.2 Problem 3 Model Development, Solution, and Prediction

Research in meteorology and hydrology indicates that systematic analysis of long-term

precipitation patterns is a key element of extreme meteorological disaster early warning. This problem first counts precipitation data for all grid points within mainland China over the past 30 years, performs data analysis to identify extreme precipitation events, defines the top 5% of precipitation values as potentially disaster-causing precipitation events, and sequentially extracts the extreme precipitation disaster dataset.

Next, to explore the combined effects of rainfall, terrain, and land use on storm disaster formation, typical geographic factors with high explanatory power are extracted from the normalized dataset to establish disaster gestation evaluation indicators for the three geographic elements of rainfall, terrain, and land use. For precipitation factors—including short-term rainfall intensity, daily precipitation range, daily mean precipitation, and precipitation duration—daily mean precipitation is selected as the precipitation disaster gestation factor based on dataset characteristics. For terrain representation factors, latitude, longitude, and elevation data of unit samples in the terrain dataset are extracted to calculate the terrain disaster gestation factor. For land use type representation factors, based on hydrological and soil science research, a hydrological model (Kostiakov infiltration model) combined with measured data from literature is used to quantify the soil moisture infiltration performance of each land use type, assessing the probability of flood disasters caused by short-duration intense rainfall and sustained heavy rainfall, thereby assigning weights to corresponding land use types and obtaining land-use coefficients. The Kostiakov infiltration model calculation formula is as follows:

$$I(t) = at^{-b}$$

In this formula: I(t) is the infiltration rate (mm/min); t is the infiltration time (min); a and b are model parameters that vary with the soil and hydrological properties corresponding to different land use types. The soil moisture infiltration curve is approximately as shown below:

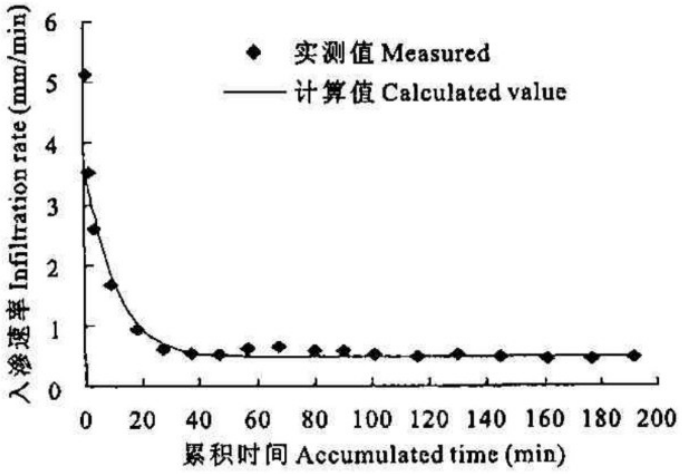


Figure 4-20 Soil infiltration rate diagram for grassland [19]

Based on the soil moisture infiltration curve, the infiltration rate is higher during the early stage of rainfall and decreases over time. To simultaneously account for the disaster probability of both short-duration intense rainfall and sustained rainfall, integration is used to calculate the total infiltration volume of rainfall per unit area within 200 minutes for different land use types. Land use types with poorer infiltration capacity have higher disaster probability, thereby weighting different land use types and calculating the land-use coefficient.

Based on the above methods, disaster gestation factors for precipitation, terrain, and land use, as well as a storm disaster gestation index dataset, are obtained. A logistic regression model is built based on these datasets to quantify the relationship between the three disaster-causing factors and the disaster index, to determine the threshold conditions for disaster occurrence. The logistic regression model principle is as follows:

Logistic regression is a widely used statistical method for classification problems, commonly applied to binary classification research. Based on linear regression, it maps the output of a linear model to the (0,1) interval by introducing a sigmoid function, thereby obtaining the probability of an event occurring. The modeling calculation formula is:

where X is the input feature vector; θ is the model parameter vector; $P(Y=1|X)$ represents the probability that Y=1 (positive class) given X; $P(Y=0|X)=1-P(Y=1|X)$ represents the probability that Y=0 (negative class). The model parameter θ is typically solved using Maximum Likelihood Estimation (MLE). For a given dataset $\{(x_i, y_i)\}_{i=1}^{n}$, where x_i represents the input feature vector and y_i represents the corresponding label (0 or 1), the log-likelihood function is:

Substituting the expression for $P(Y=1|X)$ into the above formula and taking the derivative of $l(\theta)$, optimization algorithms such as gradient descent are used to iteratively find the optimal parameter values.

Using this model for fitting, the following expression is obtained:

where p represents the disaster event probability; E represents elevation; P represents precipitation; α represents the land-use coefficient; L represents longitude; R represents latitude. The model fitting results have high accuracy.

Therefore, to forecast areas with high vulnerability during 2025–2035 based on the above three factors, the problem can be approximated as simulating and forecasting small-scale geographic indicators. Since terrain factors barely fluctuate with other factors, and combining the forecasting model of this problem with the fitting results of Problem 2, the Random Forest model is used to learn and fit land-use coefficients and precipitation disaster gestation factors over the past 30 years to build a forecasting model for these two factors, thereby estimating the distribution of areas with high vulnerability over the next decade. The construction method and operational process of the Random Forest model are the same as in Problem 2.

The core of Random Forest lies in the "ensemble learning" strategy: ensemble learning improves prediction accuracy and robustness by combining the prediction results of multiple models. Based

on the Random Forest model combined with time-series analysis and machine learning techniques, spatio-temporal patterns in the data can be captured for prediction. Specifically, 30 years of data for each grid point are learned and future curves predicted. The main variables forecasted are precipitation and land-use coefficients.

The prediction accuracy of the Random Forest model in this problem is high for both precipitation (R^2=0.90) and land-use coefficients (R^2=0.79).

The specific Random Forest construction process is as follows:

First, data cleaning is performed, covering year, longitude, latitude, elevation, precipitation, and land-use coefficients. Data are sorted, deduplicated, and processed for missing values to ensure data quality [17].

Then, for each location (longitude and latitude), we start from the last known time point:

Last known year: t_last
Last known feature value (e.g., land-use coefficient or precipitation): X_last

For each future year t_future, the following features are created:

1. Cyclic encoding of year:

2. Differencing of the target variable:

3. Lagged values of the target variable:

4. Rolling mean of the target variable:

The StandardScaler fitted during the training phase is used to standardize the generated features:

where μ and σ are the mean and standard deviation calculated on the training data.

The standardized features are input into the trained Random Forest model:

where B is the number of trees in the Random Forest and T_β is the b-th decision tree.

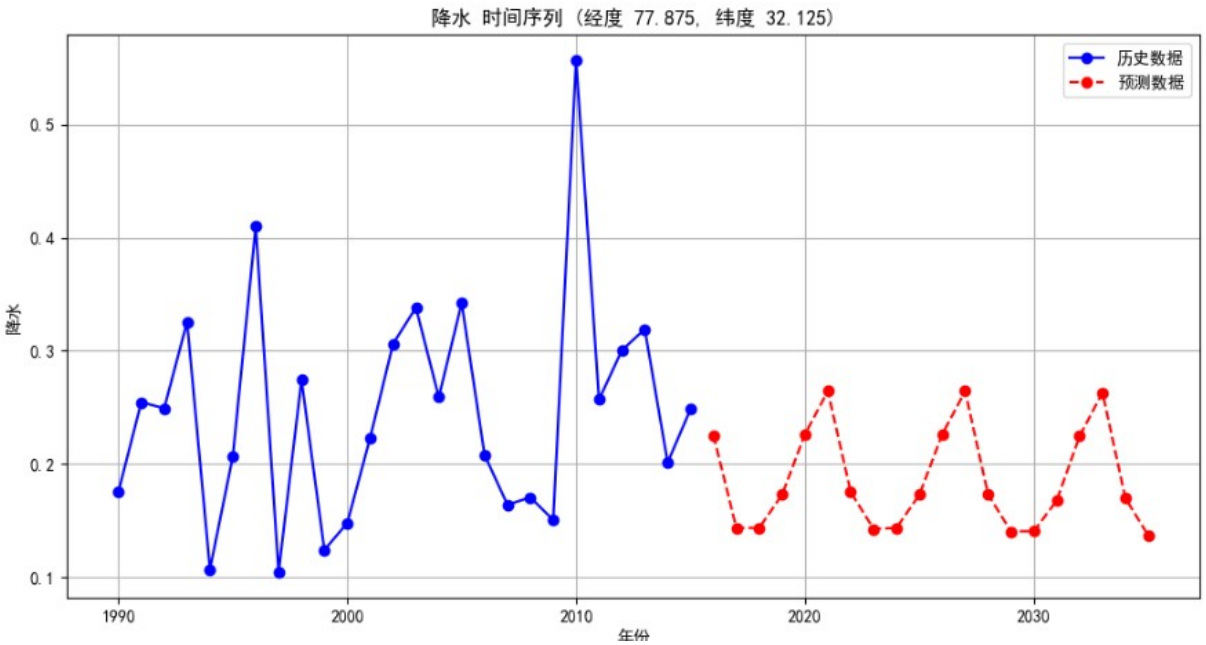
Finally, model parameters are configured as follows: number of trees: 100; maximum depth: unlimited; minimum samples per leaf node: 2.

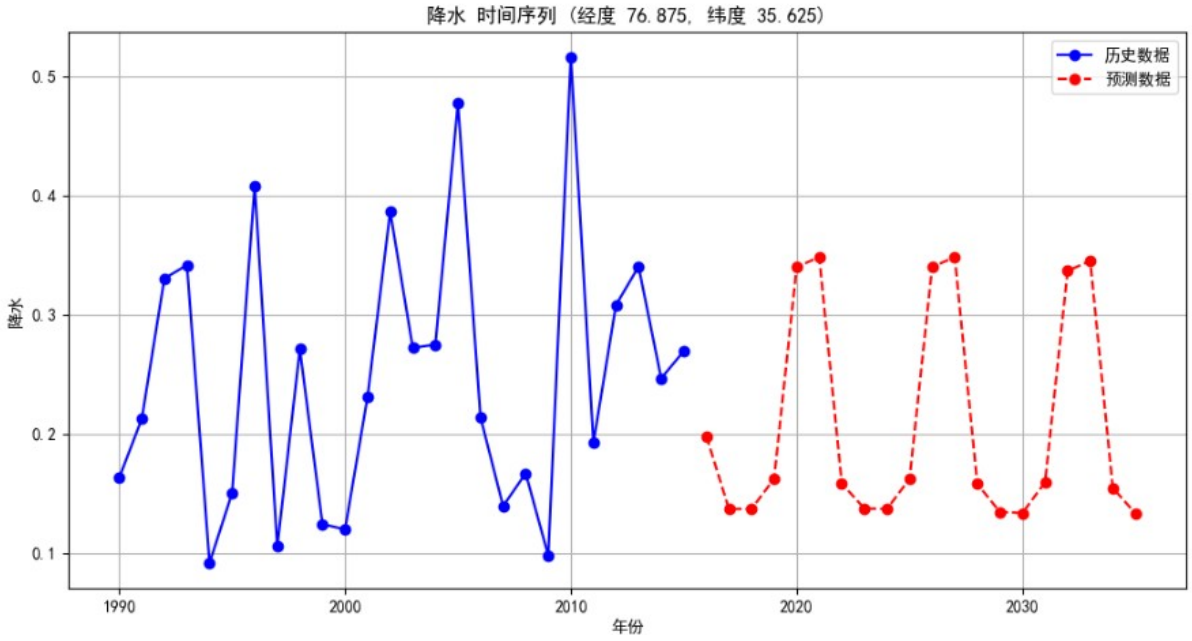
After completing the forecast for each year, the feature values are updated using the just-predicted values for the next round of feature forecasting. 、 are updated, and the process moves to the next year $t_{future+1.}$. Through this learning process, a forecasted sequence value is obtained for each location and each future year:

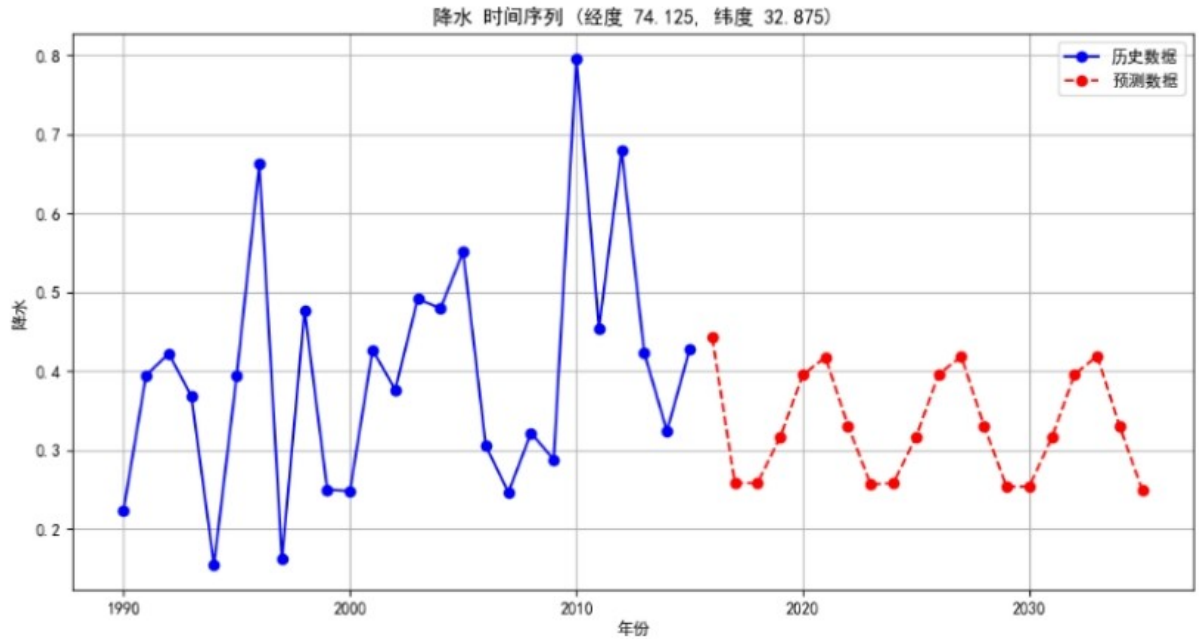
For model goodness of fit, time-series cross-validation is used to evaluate performance. R^2 scores and Mean Squared Error (MSE) are calculated for each feature (land-use coefficients and precipitation):

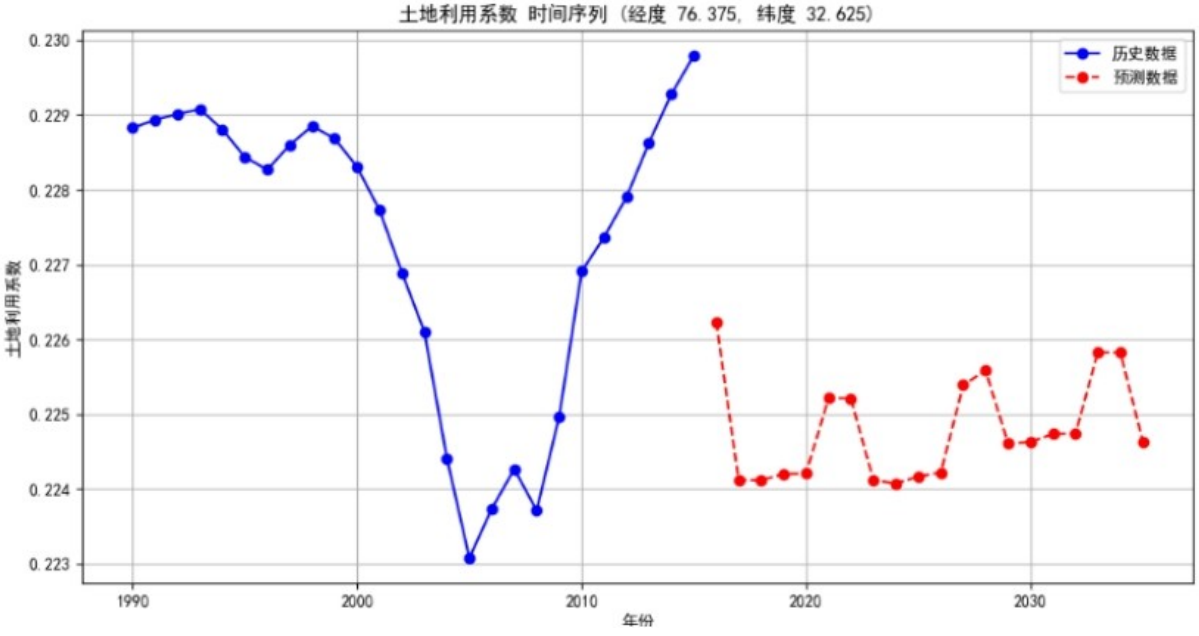
where y_i is the true value, $\hat{y}_i$ is the predicted value, and $\bar{y}$ is the mean of the true values.

The following shows the visualization analysis of model fitting and prediction results for some grid points in this dataset, where the Random Forest model learns from the change trends of the past 35 years and predicts possible changes over the next 20 years. Only some grid points are selected to display their learning curves [20]; fluctuations at different locations can be observed:


降水 时间序列（经度 77.875，纬度 32.125）
历史数据
预测数据
降水
年份
0.1
0.2
0.3
0.4
0.5
1990
2000
2010
2020
2030


降水 时间序列（经度 76.875，纬度 35.625）
历史数据
预测数据
降水
年份
0.1
0.2
0.3
0.4
0.5
1990
2000
2010
2020
2030


降水 时间序列（经度 74.125，纬度 32.875）
历史数据
预测数据
降水
年份
0.8
0.7
0.6
0.5
0.4
0.3
0.2
1990
2000
2010
2020
2030


土地利用系数 时间序列（经度 76.375，纬度 32.625）
历史数据
预测数据
土地利用系数
年份
0.230
0.229
0.228
0.227
0.226
0.225
0.224
0.223
1990
2000
2010
2020
2030

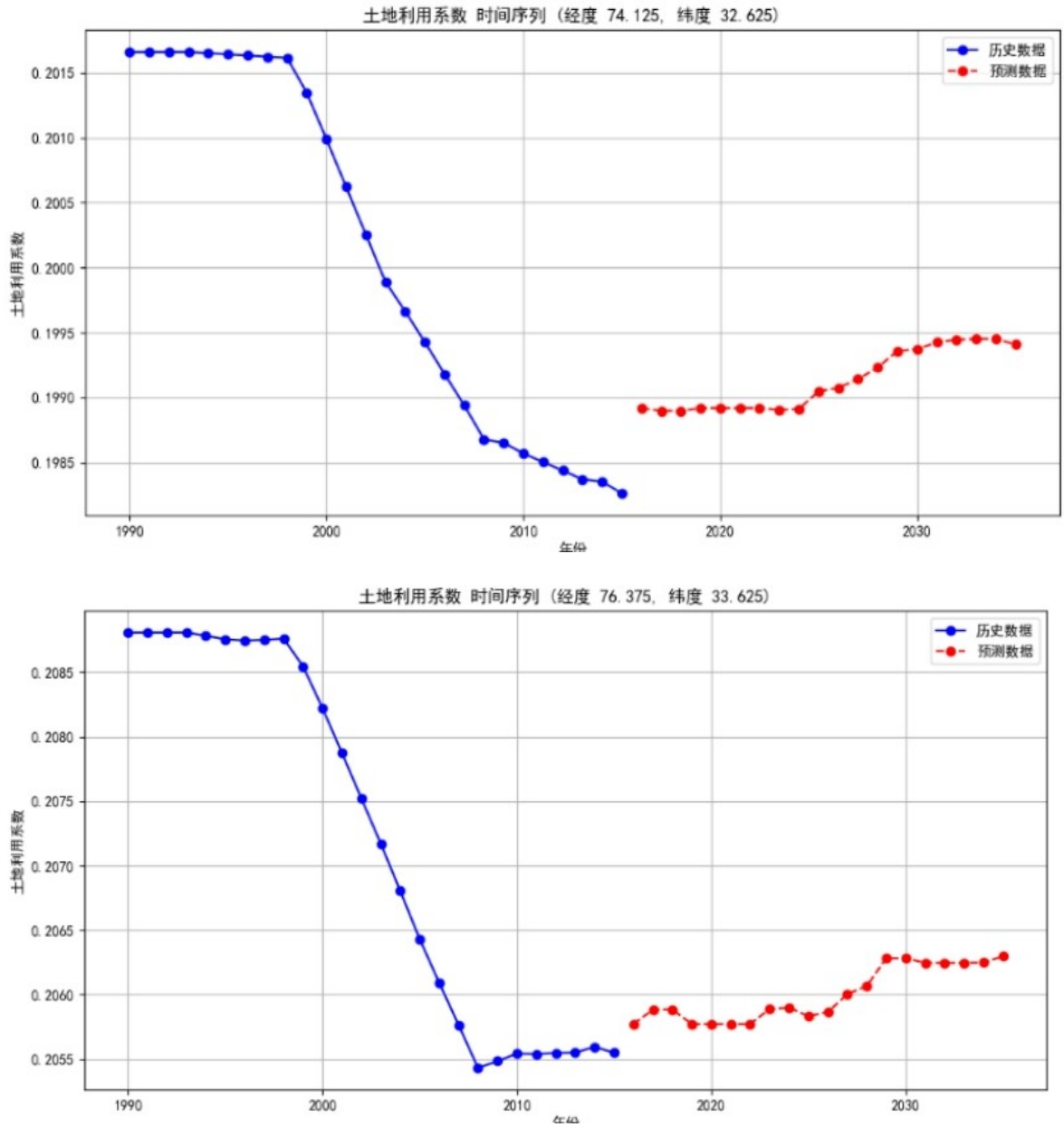


Figure 4-21 Random Forest predictions for different grid points

Note: Blue lines represent historical data; red lines represent predicted data. Precipitation and land-use coefficients are predicted for three selected grid points respectively.

Based on the Random Forest prediction results and the forecasting model, forecast data for all grid points from 2025 to 2035 are obtained. To present the forecast results, three years—2026, 2030, and 2035—are selected for description. The extreme precipitation disaster probability distribution charts (i.e., vulnerability index) are shown here. A higher index indicates a higher probability of disaster and greater vulnerability to disasters in that area. Based on the predicted 10-year data, the locations of future storm disaster points are identified, as shown below:

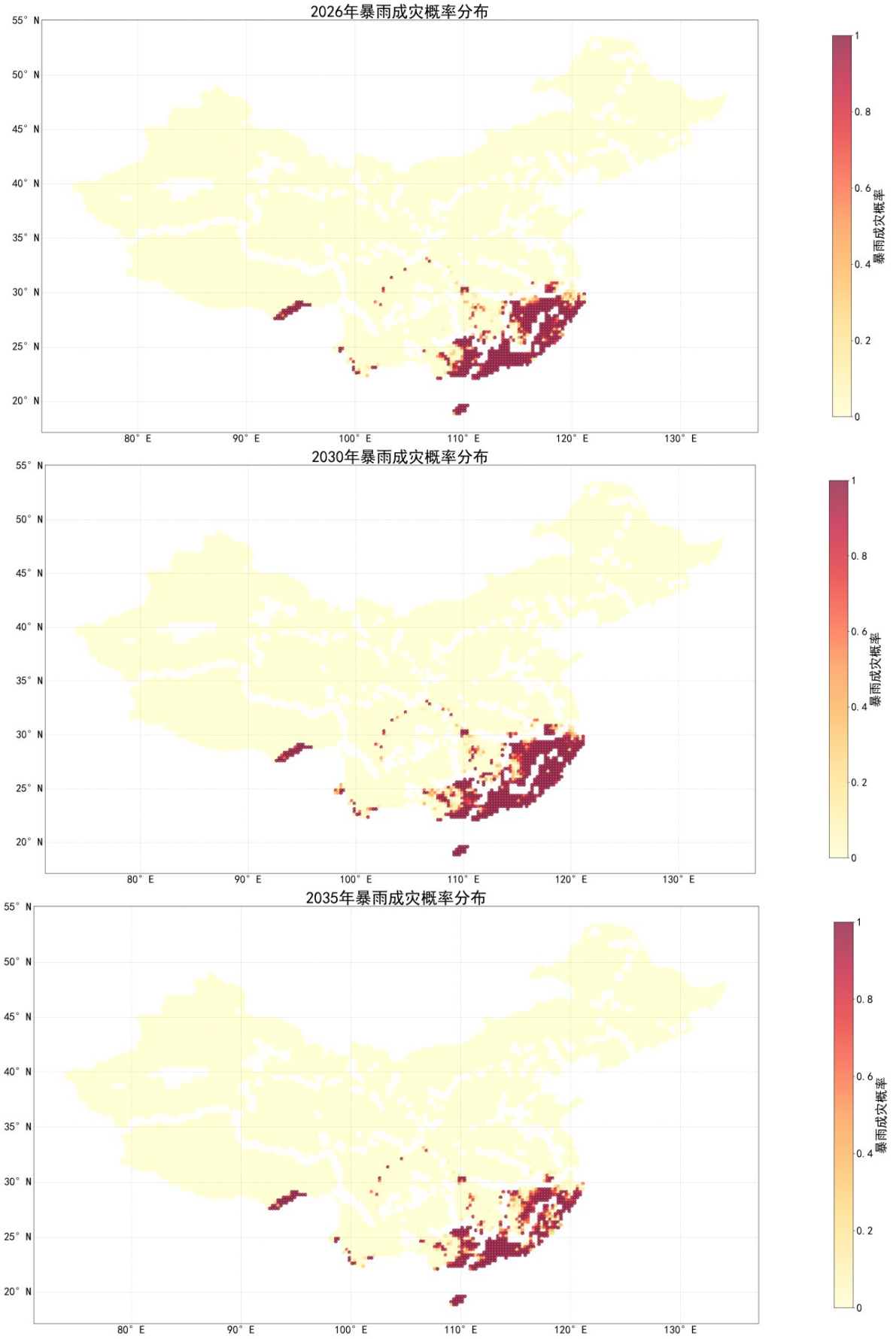


Figure 4-22 Forecast maps of storm disaster-prone areas in China for 2026, 2030, and 2035

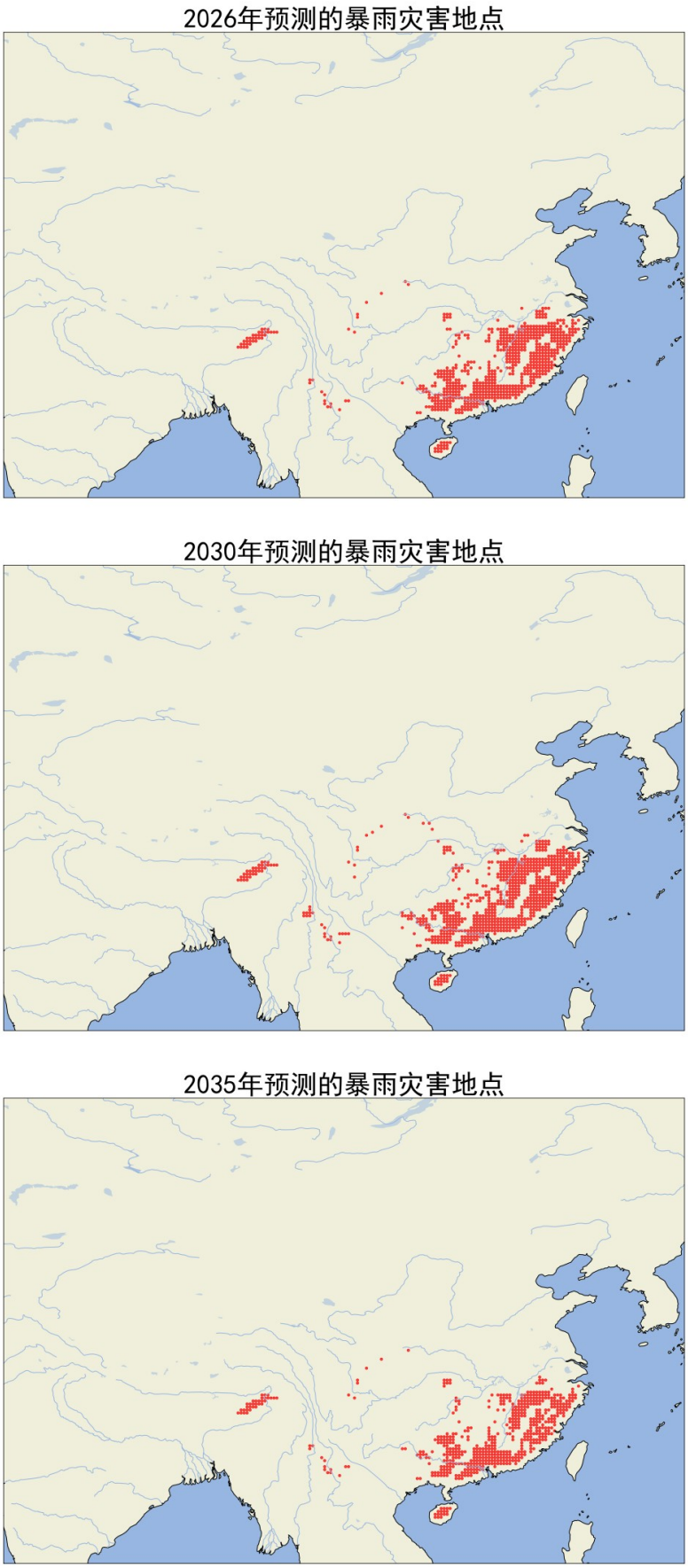
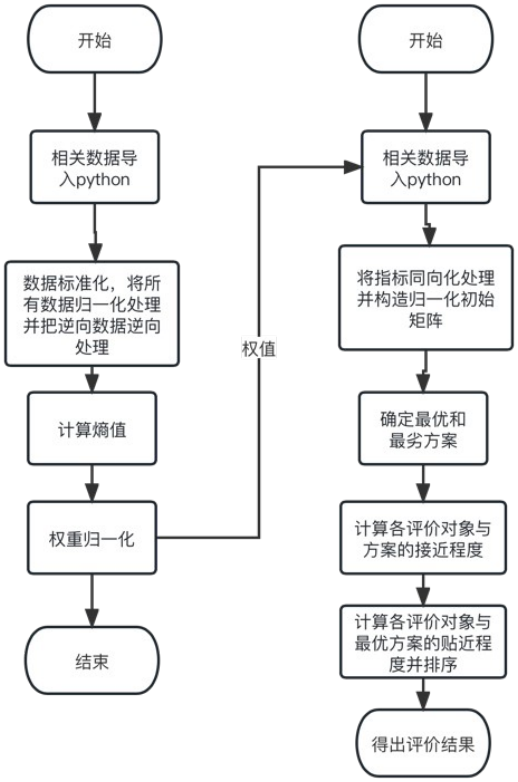


Figure 4-22 Forecast maps of storm disaster locations in China for 2026, 2030, and 2035

4.4 Problem 4: Model Development and Solution

4.4.1 Problem 4 Solution Framework



As determined from problem analysis, this problem aims to use the constructed models—combined with physical-geographic parameters (e.g., temperature, elevation, precipitation) and human-geographic parameters (e.g., GDP, population)—to explore land use type transitions and their characteristics. The discussion area is divided into four sub-regions using the Qinling-Huaihe Line and the "Hu Huanyong Line" as boundaries. The approximate starting and ending coordinates of the Qinling-Huaihe Line are (105°E, 34°N) to (120°E, 32°N); the Hu Huanyong Line runs approximately from (127°E, 50°N) to (98°E, 25°N).

This problem requires building a model describing "the characteristics and structure of China's land use change," where describing the influences of physical-geographic and human-geographic parameters on land use change is the fundamental condition. Land use types in the dataset are divided into five categories: arable land, forest, wetland, grassland, and shrubland.

4.4.2 Problem 4 Model Construction and Solution

(1) Data Cleaning

The cleaning results are detailed in Problem 2. In brief, different datasets are matched coordinate-by-coordinate and time-by-time to facilitate subsequent processing.

(2) Data Description

After cleaning, the data obtained are:

a. Temperature

where x, y are the corresponding coordinate points and t is the corresponding time.

b. Elevation (Altitude)

where x, y are the corresponding coordinate points.

c. Gross Domestic Product

where x, y are the corresponding coordinate points and t is the corresponding time.

d. Population Count

where x, y are the corresponding coordinate points and t is the corresponding time.

e. Land Use Type Proportion

where i is the corresponding land use type; x, y are the corresponding coordinate points; and t is the corresponding time.

(3) Indicator Calculation

(Note: Indicators for which no formula is given are data obtained directly from the appendix tables after cleaning.)

Per-capita land holding: the land use type proportion is multiplied by the corresponding land area to obtain the area of the corresponding type of land, and this area is then divided by the corresponding population to obtain the corresponding per-capita holding:

where L is the per-capita holding of the corresponding land type; Z is the land use type proportion; S is the unit area; and Pop is the population count.

When calculating according to this problem's solution approach for the four divided sub-regions, the corresponding per-capita holding for each part is:

It is easy to see from the above formula that S is the unit area of the cell where each data point is located, and is a constant. To simplify the formula, S is first ignored to obtain the per-capita land ratio coefficient used:

In this problem, since multiple land types exist, the following modeling analysis process uses arable land as the representative; the calculation process for other land types is the same and will not be repeated. After selecting arable land, the subsequent four proportions can be combined into the greenland proportion—meaning the sum of the greenland proportion and the arable land proportion always equals 1. Therefore, after determining arable land, the change and influence of greenland can be easily obtained.

After determining indicators that effectively describe the characteristics of temperature, population, rainfall, GDP, and per-capita arable land ratio coefficient, an evaluation model is constructed. The entropy weight method is used to calculate the weight of each indicator, and multiple linear regression is applied for fitting. A Random Forest model is built to reflect the influence of various indicators on the per-capita arable land ratio coefficient. At the same time, a TOPSIS-based multi-indicator evaluation model is established; the resulting score reflects the comprehensive changes in the per-capita arable land ratio coefficient in relation to temperature, population, rainfall, and GDP within the region. A higher score generally indicates better regional development. The main model principles used are as follows:

The basic idea of the Entropy Weight Method (EWM) is to determine objective weights based on the magnitude of indicator variability. Generally, if a certain indicator has a smaller information entropy, it indicates greater variability in the indicator values, more information provided, a greater role in comprehensive evaluation, and therefore a larger weight. Conversely, if a certain indicator has a larger information entropy, it indicates smaller variability, less information provided, a smaller role in comprehensive evaluation, and therefore a smaller weight.

The TOPSIS (Technique for Order Preference by Similarity to Ideal Solution) method is a commonly used within-group comprehensive evaluation method that can fully utilize the information from the original data; its results can precisely reflect the differences among the evaluation schemes. The basic process is: based on a normalized original data matrix, the cosine method is used to find the optimal and worst solutions among the finite alternatives. The distances between each evaluation object and the optimal and worst solutions are then calculated respectively, and the relative closeness of each evaluation object to the optimal solution is obtained as the basis for evaluating superiority and inferiority.

The principles and application background of the multiple linear regression model and Random Forest have been introduced in Problems 2 and 3 and will not be repeated here. The corresponding variables in the models can be replaced for use.

Table 4-23 Spearman Correlation Coefficients Between Indicators and Per-Capita Arable Land Ratio Coefficient by Sub-Region

Sub-Region	ln(Population)	ln(GDP)	Elevation	Temperature
Northeast	-0.6014	-0.5794	0.1220	-0.3779
Northwest	0.5430	0.4997	-0.5073	0.5675
Southeast	0.2806	0.2196	-0.3209	0.1204
Southwest	0.5765	0.5600	-0.1421	-0.0674

In actual operations, to ensure the rationality of dividing the entire mainland China into four sub-study areas, Spearman correlation coefficients were first calculated within each region using available data. As shown in the table above, the correlation coefficients in each region show a high degree of differentiation. Therefore, dividing the analysis area into these four sub-study regions is necessary to some extent. Precipitation data were not included in the simulation (same hereafter). Although precipitation is a very important independent variable as a natural factor in the study, for the purpose of simplifying the model and ensuring model rationality, the binary linear regression on precipitation from Problem 2 is used as a substitute. Specifically, the regression equation between precipitation, elevation, and temperature obtained in solving Problem 2 replaces the independent variable precipitation, thereby achieving the effect of simplifying the model.

It should be noted that the sub-region names used in the following explanations—"Northeast Region," "Northwest Region," etc.—differ from the corresponding regions as conventionally understood. In this study, the sub-region division is formed by two intersecting lines: the "Qinling-Huaihe" line and the "Hu Huanyong Line."

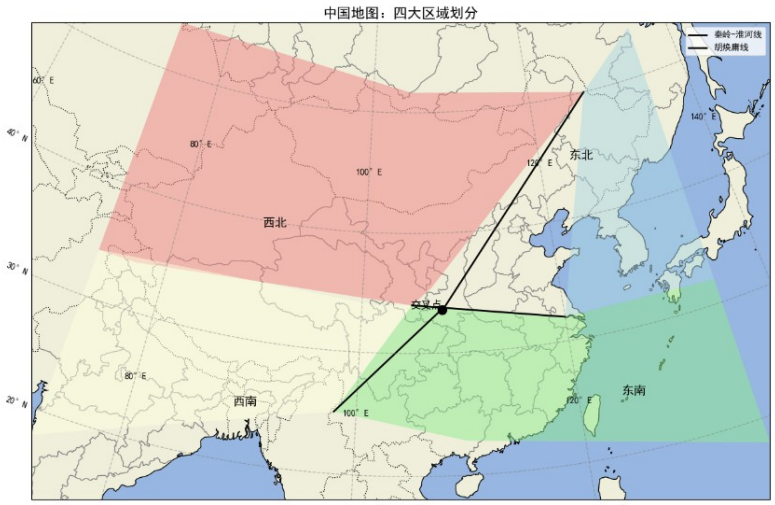


Figure 4-24 Study Area Sub-Region Division Map

Table 4-25 Summary of Entropy Weight Method Results for Northeast Region

Item	**Information Entropy Value (e)**	**Information Utility Value (d)**	**Weight Coefficient (w)**
Population (Log)	0.9808	0.0192	25.21%

GDP (Log)	0.9803	0.0197	25.83%
Elevation	0.9840	0.0160	20.91%
Temperature	0.9786	0.0214	28.05%
Per Capita Arable Land Proportion Coefficient	1.0000	0.0000	0.00%

Table 4-26 TOPSIS Evaluation Results for Northeast Region

Year	Distance to Positive Ideal Solution (D+)	Distance to Negative Ideal Solution (D-)	Relative Closeness (C)	Ranking
2005	0.0084	0.0147	0.6466	1
2006	0.0149	0.0135	0.4761	3
2007	0.0149	0.0134	0.4750	8
2008	0.0150	0.0135	0.4756	4
2009	0.0149	0.0134	0.4753	6
2010	0.0147	0.0133	0.4752	7
2011	0.0150	0.0133	0.4708	11
2012	0.0150	0.0133	0.4715	10
2013	0.0148	0.0133	0.4739	9
2014	0.0149	0.0134	0.4755	5
2015	0.0145	0.0133	0.4789	2

The information entropy value represents the degree of uncertainty of the indicator; the information utility value represents the importance of the indicator; and the weight coefficient represents the indicator's proportion in the comprehensive evaluation. Based on the weight coefficients, although GDP has the highest weight in the Northeast region's comprehensive evaluation, the weights of other items are also relatively high, with an overall uniform weight distribution. Over the past period, although GDP growth has been continuously improving and population growing, the impacts of urbanization and industrialization on the ecological environment have become increasingly severe. Therefore, the region's comprehensive development cannot be determined through simple linear proportions. Under global warming, although temperatures in the inherently cold Northeast have risen, this has also led to increased extreme climate events. The Northeast region's development index is therefore a complex coupling. Moreover, indicators do not show unidirectional change over the analyzed period; in the latter half of the analysis period, due to strong national support for ecological environment construction, improved ecological conditions also caused changes in the development index. It is worth noting that this study represents a personal perspective under the current development concept, not the development concept of the analyzed era—it is looking back at the past and

forecasting the future from today's perspective. Additionally, attention should be paid to the key role of the per-capita arable land ratio coefficient in the Northeast region. The Northeast, encompassing China's Northeast Plain and North China Plain, is an important grain-producing area and has made important contributions to ensuring national food security. The region's GDP is also highly correlated with agriculture, which is unmatched by other regions. Therefore, in comprehensive evaluation, the weight of the per-capita arable land ratio coefficient should be increased to better protect the arable land red line and national food security. These weight coefficients can be applied in comprehensive evaluation and decision analysis to help make rational decisions.

Based on the TOPSIS evaluation results, D+ does not change uniformly with increasing year; D− does not gradually increase; C does not change linearly; and the ranking results show a trend of higher in the middle and lower at both ends. This may imply that over time, the evaluated development index first experienced a gradual decline followed by a gradual rise, consistent with real-world evaluations.

Table 4-27 Summary of Entropy Weight Method Results for Northwest Region

Item	Information Entropy Value (e)	Information Utility Value (d)	Weight Coefficient (w)
Population (Log)	0.9810	0.0190	27.40%
GDP (Log)	0.9797	0.0203	29.31%
Elevation	0.9979	0.0021	3.05%
Temperature	0.9814	0.0186	26.89%
Per Capita Arable Land Proportion Coefficient	0.9908	0.0092	13.35%

Table 4-28 TOPSIS Evaluation Results for Northwest Region

Year	Distance to Positive Ideal Solution (D+)	Distance to Negative Ideal Solution (D-)	Relative Closeness (C)	Ranking
2005	0.0087	0.0135	0.6105	1
2006	0.0146	0.0124	0.4603	6
2007	0.0144	0.0125	0.4647	3
2008	0.0145	0.0126	0.4652	2
2009	0.0145	0.0124	0.4601	7
2010	0.0145	0.0122	0.4575	9
2011	0.0146	0.0122	0.4562	10

2012	0.0148	0.0122	0.4530	11
2013	0.0145	0.0122	0.4582	8
2014	0.0145	0.0124	0.4622	5
2015	0.0142	0.0123	0.4634	4

In the Northwest region, the influences of natural factors should be given more consideration. The relatively harsh natural conditions of the Northwest are the main reason for difficulties in development there. With the implementation of the Western Development Strategy, although the Northwest economy has maintained relatively rapid development, the impacts on the ecological environment were not adequately considered at the outset. Meanwhile, with climate change, extreme climate events have occurred frequently during the study period: dry areas have become drier and flood-prone areas more flood-prone. Global warming has further exacerbated this climate change, causing the originally arid inland Northwest to experience deeper drought and greater temperature differentials. What requires even more attention is that the arable land index in this region has no corresponding weight, which may be directly related to the small amount of arable land in the Northwest. This proves that under the influence of poor climate and topographic conditions in the Northwest, agriculture is not the solution to development there. In fact, in the Northwest, due to small total population and scarce water resources, the path to development should lie in areas less affected by climate conditions, such as high-end manufacturing, while also focusing on protecting and rationally transforming the natural environment. Therefore, in actual development evaluation, some ecological environment factor indicators and indicators for non-environment-dependent industries can be appropriately added.

Based on the TOPSIS evaluation results, the same pattern holds: D+, D−, and C do not change linearly; rankings show a trend of higher in the middle and lower at both ends, consistent with real-world evaluations.

Table 4-29 Summary of Entropy Weight Method Results for Southeast Region

Item	**Information Entropy Value (e)**	**Information Utility Value (d)**	**Weight Coefficient (w)**
Population (Log)	0.9810	0.0190	27.40%
GDP (Log)	0.9797	0.0203	29.31%
Elevation	0.9979	0.0021	3.05%
Temperature	0.9814	0.0186	26.89%
Per Capita Arable Land Proportion Coefficient	0.9908	0.0092	13.35%

Table 4-30 TOPSIS Evaluation Results for Southeast Region

Year	**Distance to**	**Distance to**	**Relative**	**Ranking**

	Positive Ideal Solution (D+)	Negative Ideal Solution (D-)	Closeness (C)	
2005	0.0087	0.0135	0.6105	1
2006	0.0146	0.0124	0.4603	6
2007	0.0144	0.0125	0.4647	3
2008	0.0145	0.0126	0.4652	2
2009	0.0145	0.0124	0.4601	7
2010	0.0145	0.0122	0.4575	9
2011	0.0146	0.0122	0.4562	10
2012	0.0148	0.0122	0.4530	11
2013	0.0145	0.0122	0.4582	8
2014	0.0145	0.0124	0.4622	5
2015	0.0142	0.0123	0.4634	4

The Southeast region encompasses most of China's economically strong provinces and has a relatively low average elevation; consequently, elevation is not an indicator affecting the development index here. The Southeast has a very dense population, with most areas having a subtropical monsoon climate. Compared with other regions, both rainfall and heat conditions are more favorable for crop growth, so more crops can be harvested per unit area, which led to a smaller contribution of the per-capita arable land ratio coefficient to the overall evaluation system. In addition, this region's rapid economic development relies mainly on light industry and the service sector, with the primary industry accounting for a relatively small proportion. It can therefore be seen that the Southeast clearly shows an increased proportion of human-geographic indices, with GDP and population having higher weights overall. In evaluation, the influence of human-geographic indicators should therefore be given greater importance, while the effects of temperature and rainfall on this region must also be carefully noted.

Based on the TOPSIS evaluation results, the same pattern holds, consistent with real-world evaluations.

Table 4-31 Summary of Entropy Weight Method Results for Southwest Region

Item	Information Entropy Value (e)	Information Utility Value (d)	Weight Coefficient (w)
Population (Log)	0.9725	0.0275	22.48%
GDP (Log)	0.9697	0.0303	24.78%
Elevation	0.9673	0.0327	26.79%
Temperature	0.9737	0.0263	21.53%
Per Capita Arable	0.9946	0.0054	4.42%

Land Proportion Coefficient				

Table 4-32 TOPSIS Evaluation Results for Southwest Region

Year	Distance to Positive Ideal Solution (D+)	Distance to Negative Ideal Solution (D-)	Relative Closeness (C)	Ranking
2005	0.0091	0.0139	0.6066	1
2006	0.0150	0.0128	0.4607	5
2007	0.0149	0.0128	0.4610	4
2008	0.0150	0.0128	0.4624	3
2009	0.0149	0.0127	0.4603	6
2010	0.0148	0.0126	0.4594	8
2011	0.0150	0.0125	0.4557	10
2012	0.0151	0.0126	0.4554	11
2013	0.0148	0.0126	0.4594	9
2014	0.0149	0.0127	0.4603	7
2015	0.0146	0.0126	0.4632	2

The distribution of the Southwest region is very similar to that of the Northwest region, which may be due to the similarity between the human and natural geographic indicators of the actual Southwest and Northwest regions. The Northwest is mostly arid desert, while the Southwest is also a high-altitude plateau and mountain area unfavorable for human habitation. In fact, according to the geographic significance of the Qinling-Huaihe Line, the constraint on the western region is not obvious. In other words, in future evaluations, it may be possible to combine the Northwest and Southwest regions for joint analysis.

Based on the TOPSIS evaluation results, the same pattern holds, consistent with real-world evaluations.

The main task of this problem is to determine the transitions of land use types and their characteristics. As a simple start, multiple linear regression is used for a simple fitting.

General multiple linear regression is described as:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_k X_k + \varepsilon$$

In this formula, Y is the dependent variable; X_m are the independent variables; θ_0 is the constant term; θ_m are the regression coefficients for the corresponding independent variables; and ε is the error term.

In this study, the dependent variable is the per-capita arable land ratio coefficient, and the independent variables are precipitation, temperature, elevation, GDP, and population. Precipitation is quantified by the binary linear regression result from Problem 2. Therefore, in practice, the independent variables are only temperature, elevation, GDP, and population (the regression coefficient for the rainfall column below is included to more intuitively show the influence of precipitation—a common climate indicator—on the per-capita arable land ratio coefficient).

Table 4-33 Multiple Linear Regression Fitting Results for Northeast Region

Variable	coef	std err	t	P>|t|	[0.025	0.975]
Constant	0.001149	4.33E-06	265.6823	0	0.001141	0.001158
Elevation	-0.00043	1.2E-05	-36.252	2.9E-269	-0.00046	-0.00041
Temperature	-0.00017	1.25E-05	-13.3831	1.8E-40	-0.00019	-0.00014
Precipitation	-0.00014	6.47E-06	-21.2253	1.14E-97	-0.00015	-0.00012
Population (Log)	-0.00052	9.56E-06	-54.0088	0	-0.00054	-0.0005
GDP (Log)	-0.0001	7.92E-06	-12.8088	3.02E-37	-0.00012	-8.6E-05

The parameters for the Northeast region obtained through multiple linear regression are shown in the above table. All five independent variables show negative correlation coefficients, but the values are small—indicating that the per-capita arable land ratio coefficient in the Northeast is relatively less influenced by changes in the independent variables. This is consistent with the Northeast's role as an important grain-producing area; the stability of arable land in the Northeast directly affects China's food security. Therefore, under the influence of policy, arable land area in the Northeast changes relatively little with other factors.

Table 4-34 Multiple Linear Regression Fitting Results for Northwest Region

Variable	coef	std err	t	P>|t|	[0.025	0.975]
Constant	0.013819	0.006651	2.077925	0.03772	0.000784	0.026854
Elevation	-0.00064	0.024452	-0.02618	0.979111	-0.04857	0.047287
Temperature	0.032654	0.023682	1.378873	0.167939	-0.01376	0.079071
Precipitation	-0.01465	0.010067	-1.45501	0.14567	-0.03438	0.005084

Population (Log)	-0.03387	0.026244	-1.29066	0.196826	-0.08531	0.017566
GDP (Log)	-0.00283	0.025059	-0.1131	0.909952	-0.05195	0.046281

The parameters for the Northwest region are shown in the above table. Temperature has a positive influence on the per-capita arable land ratio coefficient, while the rest have negative influences. This may be due to urbanization: the Northwest region tends to develop areas with more precipitation and lower elevation into cities, or such areas become greenland due to their relatively favorable natural conditions for environmental improvement, while arable land relies more on artificial irrigation and human reclamation.

Table 4-35 Multiple Linear Regression Fitting Results for Southeast Region

Variable	**coef**	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
Constant	0.003926	2.88E-05	136.3421	0	0.00387	0.003982
Elevation	-0.00019	2.96E-05	-6.40975	1.48E-10	-0.00025	-0.00013
Temperature	-8.6E-06	4.13E-05	-0.20792	0.835295	-9E-05	7.23E-05
Precipitation	-0.00012	3.8E-05	-3.1978	0.001386	-0.0002	-4.7E-05
Population (Log)	0.000873	6.3E-05	13.85577	1.62E-43	0.00075	0.000997
GDP (Log)	-0.00087	6.23E-05	-13.992	2.44E-44	-0.00099	-0.00075

The parameters for the Southeast region are shown in the above table. The Southeast encompasses most of China's economically strong provinces. Analysis of the regression coefficients shows that many indicators have a significant influence on the per-capita arable land ratio coefficient; only population shows a weak positive correlation, while all others show negative correlations. In the economically developed Southeast, agricultural development is relatively mature, and due to favorable rainfall and heat conditions, the demand for per-capita arable land ratio coefficient is lower than in other regions. Therefore, the per-capita arable land ratio coefficient may remain in a relatively stable state for a considerable period.

Table 4-36 Multiple Linear Regression Fitting Results for Southwest Region

Variable	**coef**	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
Constant	0.00259	3.36E-05	77.15059	0	0.002524	0.002655
Elevation	-0.0012	6.03E-05	-19.8631	3.12E-87	-0.00132	-0.00108
Temperature	-0.00033	7.39E-05	-4.53027	5.91E-06	-0.00048	-0.00019
Precipitation	0.000255	5.12E-05	4.985286	6.22E-07	0.000155	0.000356

Population (Log)	1.91E-05	6.88E-05	0.277433	0.78145	-0.00012	0.000154
GDP (Log)	0.000396	6.9E-05	5.732833	9.97E-09	0.00026	0.000531

The parameters for the Southwest region are shown in the above table. The Southwest is significantly influenced by elevation; the relationship between the per-capita arable land ratio and natural indicators meets expectations: areas with lower elevation, suitable temperature, and slightly more precipitation are more likely to develop into arable land. GDP shows a positive correlation with per-capita arable land ratio, possibly indicating that in this region, the contribution from primary agriculture accounts for a larger proportion of GDP.

Although the multiple regression model can reasonably explain the relationship between per-capita arable land ratio and physical-geographic and human-geographic indicators, some fits' R^2 are not satisfactory. To better optimize the model, Random Forest is used for fitting. The selected variables are the same as those used in the multiple regression above and will not be repeated.

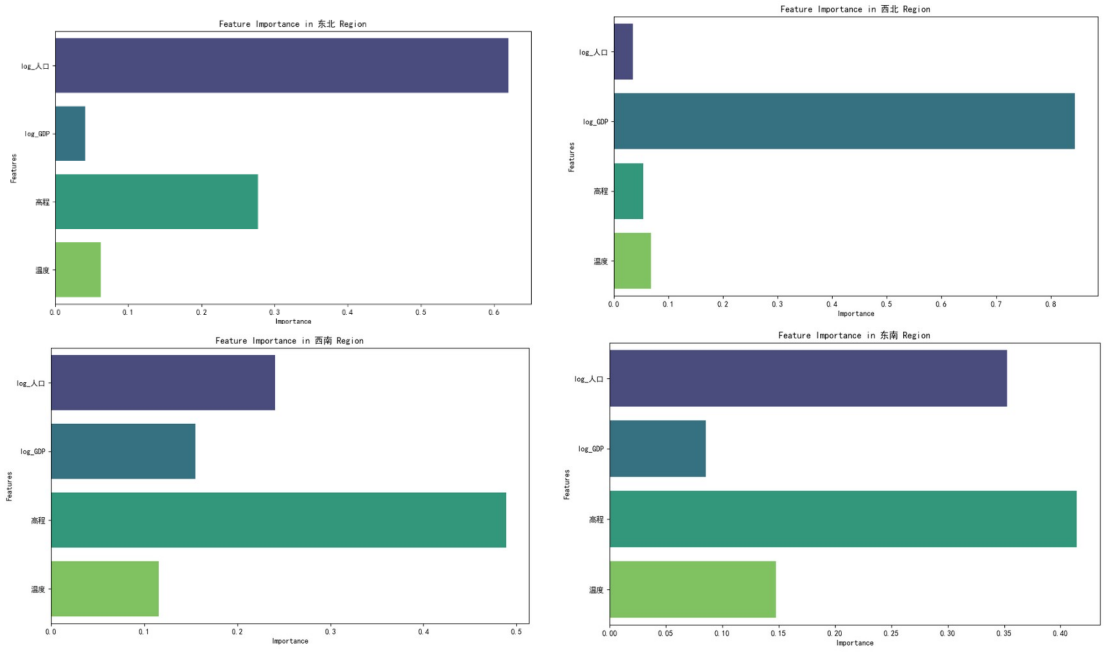


Figure 4-37 Random Forest fitting weight chart for four sub-regions

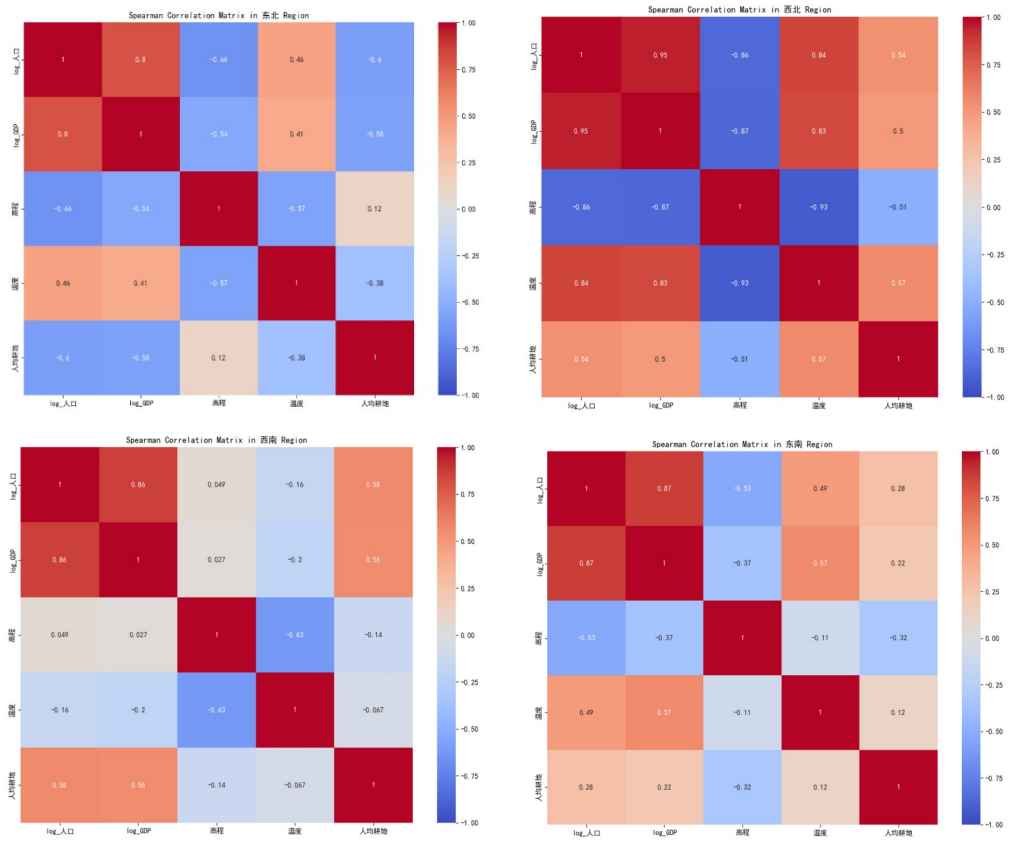


Figure 4-38 Spearman correlation coefficient chart of Random Forest fitting for four sub-regions

The Random Forest simulation results for the four regions are shown above.

Northeast Region: Spearman correlation coefficients: Population (-0.6014) and GDP (-0.5794) show a moderate negative correlation with per-capita arable land, possibly indicating that urbanization and economic development in the Northeast may lead to a decrease in per-capita arable land. Elevation (0.1220) shows a very weak positive correlation. Temperature (-0.3779) shows a weak to moderate negative correlation; the Northeast has a large climate span with marked seasonal temperature differentials. Arable land is mainly distributed in the northern part of the Northeast region, contributing to this result.

Random Forest feature importance: Population (0.6197) is the most important feature, emphasizing the dominant role of the population factor in arable land distribution in the Northeast. Elevation (0.2773) is the second most important feature. GDP and temperature are of relatively lower importance.

Northwest Region: Spearman correlation coefficients: Population (0.5430) and GDP (0.4997) show a moderate positive correlation with per-capita arable land, possibly indicating that economic and population development in the Northwest has a positive association with arable land use. Elevation (-0.5073) shows a moderate negative correlation, indicating that higher elevation may mean less per-capita arable land. Temperature (0.5675) shows a moderate positive correlation, suggesting that warmer areas may have more per-capita arable land.

Random Forest feature importance: GDP (0.8441) is the most important feature, emphasizing the critical role of economic development in arable land distribution in the Northwest. Temperature (0.0676) is the second most important feature, again emphasizing the importance of climate conditions. Population and elevation are of relatively lower importance.

Southeast Region: Spearman correlation coefficients: All factors show relatively weak correlations with per-capita arable land. Population (0.2806) and GDP (0.2196) show a weak positive correlation. Elevation (-0.3209) shows a weak negative correlation; temperature (0.1204) shows almost no correlation.

Random Forest feature importance: Elevation (0.4144) is the most important feature, possibly reflecting the influence of the complex terrain in the Southeast on arable land distribution; the Southeast has many hilly areas and arable land distribution is highly related to terrain. Population (0.3527) is the second most important feature. Temperature (0.1475) and GDP (0.0854) are of relatively lower importance.

Southwest Region: Spearman correlation coefficients: Population (0.5765) and GDP (0.5600) show a moderate positive correlation with per-capita arable land, similarly to the Northwest, possibly indicating that economic and population development has a positive association with arable land use. Elevation (-0.1421) shows a relatively weak negative correlation, indicating that higher elevation may mean less per-capita arable land. Temperature (-0.0674) shows an extremely weak positive correlation.

Random Forest feature importance: Elevation (0.4887) is the most important feature, emphasizing the important influence of terrain on arable land in the Southwest; the Southwest has a higher average elevation, and elevation is the key constraint on agricultural development. Population (0.2407) is the second most important feature. Temperature and GDP are of relatively lower importance.

The results obtained from Random Forest are similar to China's actual situation, reasonably and accurately explaining the different land use changes in each region. Northeast: Random Forest model R^2= 0.9481; Northwest: Random Forest model R^2= 0.3150; Southeast: Random Forest model R^2= 0.9008; Southwest: Random Forest model R^2= 0.7388. Except for the Northwest region, the model's explanatory power is high. As mentioned earlier, because the Qinling-Huaihe Line does not create a clear division in the western region, the Northwest and Southwest sub-regions are combined for simulation.

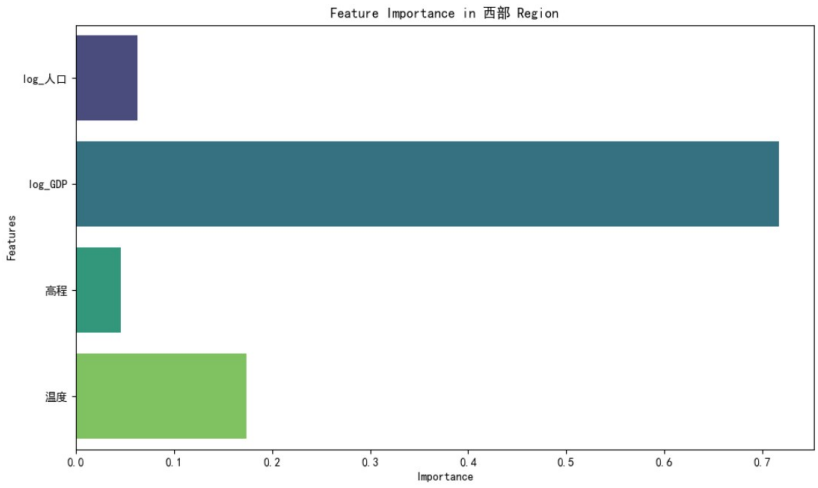


Figure 4-39 Random Forest fitting weight chart for the combined Western Region

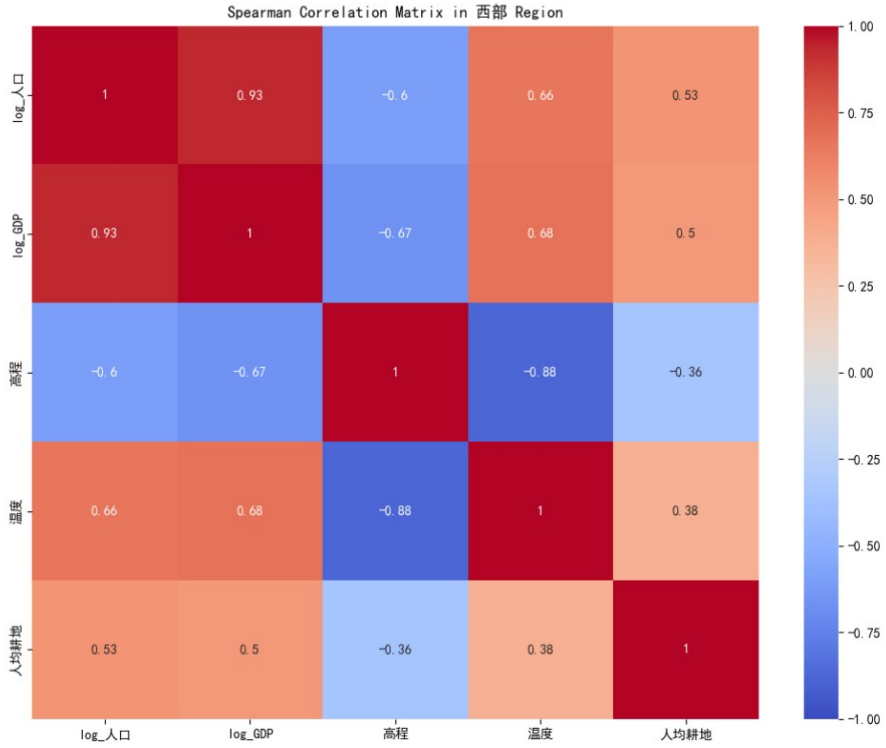


Figure 4-40 Spearman correlation coefficient chart of Random Forest fitting for the Western Region

The fitting results for the Western region are shown above. Notably, this R^2= 0.8831 is far higher than the explanatory values from the separate fittings; this classification is therefore clearly more reasonable.

Spearman correlation coefficients: Population (0.5309) and GDP (0.5023) show a moderate positive correlation with per-capita arable land, possibly indicating that economic and population development in the Western region has a positive association with arable land use. Elevation (-0.3564) shows a weak to moderate negative correlation, indicating that higher elevation may mean less per-capita arable land. Temperature (0.3783) shows a weak to moderate positive correlation, suggesting that warmer areas may have more per-capita arable land.

Random Forest feature importance: GDP (0.7167) is the most important feature, emphasizing the critical role of economic development in arable land distribution in the Western region. Temperature (0.1741) is the second most important feature. Population and elevation are of relatively lower importance.

Economic development remains the most urgent issue to be addressed in the Western region. West of the Hu Huanyong Line, human-geographic indicators play a dominant role in influencing land use change.

In fact, the influencing factors of per-capita arable land differ significantly across regions, reflecting the unique characteristics of different geographic regions in China. GDP and temperature in the Western region, population in the Northeast, and elevation and population in the Southeast are the main factors affecting per-capita arable land in their respective regions.

Correlation and feature importance do not always agree and should be further validated through additional model simulations.

Unfortunately, even when all variables obtained in this task are input, the relationships between various factors and per-capita arable land in the Southeast region remain relatively weak, possibly suggesting the existence of other **important factors not included in the analysis**.

V. Model Evaluation and Extension

5.1 Model Strengths

- This study employs a large volume of geographic big data. By leveraging data-driven analysis and prediction of the spatio-temporal evolution characteristics of the geographic environment, and by using big data learning, the practicality and reliability of the models are ensured.
- Through comprehensive analysis of multiple variables, the models integrate various factors such as rainfall, terrain, and land use, comprehensively considering their roles in the formation of extreme weather and the disaster-formation process, ensuring prediction accuracy.
- By constructing descriptive statistical methods, the models effectively provide multi-level spatio-temporal descriptions of climate, terrain, land use, and human-geographic indicators, effectively representing their evolutionary characteristics and presenting them intuitively in map form.
- Using the multiple linear regression model to first simplify complex problems prevents overfitting and inappropriate fitting methods, reducing workload while improving prediction accuracy.
- The introduction of the Gumbel distribution reasonably simulates the probability density function of precipitation and determines the threshold values for storm-induced disasters based thereon; this is more accurate and reasonable than using purely subjective judgment.

- When multiple linear regression performs poorly, the logistic regression model is decisively introduced, summarizing the corresponding response values and providing good explanatory power for the dependent variable.
- The introduction of the Random Forest model is the core of this model; through learning on large amounts of data, the Random Forest model captures key dependency weights, improves prediction accuracy, and can reasonably describe and explain the characteristics of related indicators.

5.2 Model Limitations

The accuracy of the data limits the prediction results of this model. Numerous missing values and data errors have introduced interference into model training. The provided data cannot meet the requirements for normal simulation; for example, land use lacks hard-surface attachment caused by human activities, and the inability to refine rainfall data to the hourly level means that predictions of storm-induced disasters will inevitably deviate from actual conditions.

The models are primarily trained on historical data and are very limited in terms of adaptability to future changes. From a physical geography perspective, temperature and rainfall undergo cyclical changes on longer scales—often hundreds or thousands of years—making it likely that data spanning only a few decades will fail to accurately capture the period and simulate the patterns.

Because subjective judgments are involved in the models multiple times—such as the definition of heavy rainfall and the definition of disaster formation—these subjective judgment factors will affect the applicability of the models. Different subjective thresholds will lead to different results.

During the extraction of terrain evaluation factors, relatively few factors were selected; elevation was extracted as the representative factor, but a single variable is insufficient to describe China's complex terrain and may affect fitting results. During subsequent model optimization, it is planned to extract slope and aspect as additional terrain evaluation factors.

Unfortunately, all three authors of this paper proposed using the LSTM model for prediction because LSTM is particularly suitable for processing sequence data with long-range dependencies. However, due to excessively high computational complexity—requiring large amounts of computational resources and time—it was impossible to complete the simulation within the required time given available hardware. Constrained by objective factors, this method had to be abandoned.

The actual applicability of this model may be higher than what is shown in this paper. In fact, using GIS software such as ArcMap's Geodetector for determining disaster threshold conditions and vulnerability estimation may be easier. However, since all calculations in ArcMap are packaged software functions, it is difficult to understand the mathematical models in the process, and it is uncertain whether ArcMap is applicable to this competition, as using integrated software may not demonstrate the team's model-building capabilities. Extreme high temperatures, droughts, and other phenomena can also be directly simulated by Geodetector, but due to time constraints, there was no opportunity to perform the simulations after completing the modeling process in this paper.

5.3 Model Extension

- After being trained on larger amounts of data, this model can provide recommendations and decision guidance for the government on ecological protection and disaster prevention.
- After being developed into interactive visualization operations, this model can be used for education and training purposes.
- This model can respond to climate change in a relatively significant manner and, combined with human and natural geographic parameters, can analyze natural systems and human activities.

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Appendix

Note: The code was independently structured and generated after incorporating AI debugging suggestions.

Reference for TIFF data extraction and resampling code:

https://blog.csdn.net/mm15707472721/article/details/135513655

问题一

1. 提取降水量和土地利用格点数据

```
import xarray as xr
import pandas as pd
import numpy as np

file_path = r'E:\CHM_PRE_0.25dg_19612022.nc'

try:
    with xr.open_dataset(file_path, decode_times=True) as data:
        start_year = 1990
        end_year = 1991
        yearly_mean_precip = pd.DataFrame(index=pd.date_range(start=f'{start_year}-01-01',
end=f'{end_year}-12-31'))
```

```
        for year in range(start_year, end_year + 1):
            precipitation = data['pre'].sel(time=slice(f'{year}-01-01', f'{year}-12-31'))

            precipitation = precipitation.where(precipitation != -99.9, np.nan)
            daily_mean = precipitation.mean(dim=('longitude', 'latitude'), skipna=True).values
            yearly_mean_precip.loc[f'{year}-01-01':f'{year}-12-31', str(year)] = daily_mean

        excel_file_path = r'E:\yearly_mean_precipitation_1990_2020.xlsx'
        yearly_mean_precip.to_excel(excel_file_path)

        print(f"Excel 文件已保存到 {excel_file_path}")

except Exception as e:
    print(f"Error processing data: {e}")

-------

data_path = r"E:\land"

# 获取所有 GeoTIFF 文件
geotiff_files = [f for f in os.listdir(data_path) if f.endswith('.tif')]

# 筛选 1990 到 2020 年的文件
filtered_files = [f for f in geotiff_files if '1990' <= f.split('-')[1][:4] <= '2020']

for file in filtered_files:
    file_path = os.path.join(data_path, file)
    with rasterio.open(file_path) as src:
        data = src.read(1)  # 读取第一个波段数据


        df = pd.DataFrame(data)
        output_path = os.path.join(data_path, f"{os.path.splitext(file)[0]}.xlsx")

        df.to_excel(output_path, index=False, header=False)
        print(f"数据已保存到 {output_path}")



2. 地图绘图代码
plt.rcParams['font.sans-serif'] = ['SimHei']
plt.rcParams['axes.unicode_minus'] = False

file_path = r'E:\CHM_PRE_0.25dg_19612022.nc'

# 指定处理的年份
years = [1990, 2000, 2005, 2010, 2015, 2020]

# 打开数据集并处理数据
try:
    with xr.open_dataset(file_path, decode_times=True) as data:
        fig, axs = plt.subplots(3, 2, figsize=(12, 15), subplot_kw={'projection': ccrs.PlateCarree()})
        axs = axs.flatten()  # 将 2x3 的数组展平，便于索引
```

```
    for i, year in enumerate(years):
        precipitation = data['pre'].sel(time=slice(f'{year}-01-01', f'{year}-12-31'))
        precipitation = precipitation.where(precipitation != -99.9, np.nan)
        total_precipitation = precipitation.sum(dim='time').values
        lon = data['longitude']
        lat = data['latitude']
        ax = axs[i]
        ax.add_feature(cfeature.BORDERS, linestyle='-', linewidth=1.5, edgecolor='black')

        # 绘制降水量的格点图（填充图层）
        filled_contour = ax.contourf(lon, lat, total_precipitation, transform=ccrs.PlateCarree(),
cmap='Blues')
        plt.colorbar(filled_contour, ax=ax, orientation='horizontal', label='年平均降水量 (mm)')

        # 绘制等值线（200、400、800 毫米）
        contour = ax.contour(lon, lat, total_precipitation, levels=[200, 400, 800],
                             colors=['blue', 'green', 'red'], linewidths=1, transform=ccrs.PlateCarree())

        # 添加等值线标签
        ax.clabel(contour, inline=True, fontsize=8, fmt='%d mm')

        ax.set_title(f'{year}年中国年降水量分布图', fontsize=16)

    plt.tight_layout()
    plt.savefig('E:\six-year-rain')
    plt.show()
```

3.土地利用转移矩阵计算绘图

```
land_type_mapping = {
    'cropland': '耕地',
    'wetland': '湿地',
    'grass': '草地',
    'forest': '林地',
    'shrub': '灌木'
}

def generate_colors(n):
    color_palette = [
        "#1f77b4", "#ff7f0e", "#2ca02c", "#d62728", "#9467bd",
        "#8c564b"
    ]
    return color_palette[:n]


def load_land_use_data(land_data_path, years, land_types):
    data = {year: {} for year in years}
    for year in years:
        for land_type in land_types:
            file_name = f"{land_type}-{year}.tif"
            file_path = os.path.join(land_data_path, file_name)
            if os.path.exists(file_path):
                with rasterio.open(file_path) as src:
                    data[year][land_type] = src.read(1)
            else:
                print(f"No land use data available for {land_type} in year {year}.")
```

```
    return data


def calculate_transition_matrix(data_start, data_end, land_types, pixel_area):
    transition_matrix = np.zeros((len(land_types), len(land_types)))
    for i, type_start in enumerate(land_types):
        for j, type_end in enumerate(land_types):
            if type_start in data_start and type_end in data_end:
                pixel_transition = data_start[type_start] * data_end[type_end] * pixel_area
                transition_matrix[i, j] = np.sum(pixel_transition)
    return transition_matrix

def print_transition_matrix(transition_matrix, land_types):
    print("\n 转移矩阵:")
    print("From/To".ljust(10), end="")
    for lt in land_types:
        print(f"{land_type_mapping[lt]:>10}", end="")
    print()

    for i, from_type in enumerate(land_types):
        print(f"{land_type_mapping[from_type]:<10}", end="")
        for j in range(len(land_types)):
            print(f"{transition_matrix[i, j]:10.0f}", end="")
        print()


def create_sankey_diagram(transition_matrices, years, land_types, output_file):
    labels = []
    source = []
    target = []
    value = []
    colors = generate_colors(len(land_types))

    # 为每个时间点的每种土地类型生成深色节点颜色
    node_colors = []
    for _ in years:
        for color in colors:
            rgba_color = f"rgba({int(color[1:3], 16)},{int(color[3:5], 16)},{int(color[5:7], 16)},0.8)"
            node_colors.append(rgba_color)

    for i, year in enumerate(years):
        labels.extend([f"{land_type_mapping[lt]} ({year})" for lt in land_types])

        if i < len(years) - 1:
            matrix = transition_matrices[i]
            offset = i * len(land_types)
            for from_type in range(len(land_types)):
                for to_type in range(len(land_types)):
                    source.append(offset + from_type)
                    target.append(offset + len(land_types) + to_type)
                    value.append(matrix[from_type, to_type])

    # 生成浅色连接线颜色
    link_colors = []
    for s in source:
        base_color = colors[s % len(land_types)]
        rgba_color = f"rgba({int(base_color[1:3], 16)},{int(base_color[3:5], 16)},{int(base_color[5:7],
```

```
16)},0.3)"
        link_colors.append(rgba_color)

    fig = go.Figure(data=[go.Sankey(
        node=dict(
            pad=15,
            thickness=20,
            line=dict(color="#444", width=0.5),
            label=labels,
            color=node_colors,
            font=dict(size=16, color="black")  # 增加节点标签字体大小
        ),
        link=dict(
            source=source,
            target=target,
            value=value,
            color=link_colors,
            hoverinfo="all",
            hovertemplate="从 %{source.label}<br>" +
                          "到 %{target.label}<br>" +
                          "面积: %{value:.2f}<extra></extra>"
        )
    )])

    fig.update_layout(
        title_text="土地利用转移矩阵 (1990-2019)",
        font=dict(size=18, family="Arial", color="#444"),  # 增加整体字体大小
        title_font=dict(size=24),  # 增加标题字体大小
        paper_bgcolor='rgba(0,0,0,0)',
        plot_bgcolor='rgba(0,0,0,0)',
        width=1800,  # 增加图表宽度
        height=1200  # 增加图表高度
    )
    fig.write_image(output_file)
    print(f"多时间段土地利用转移 Sankey 图已保存为 {output_file}")


# 主程序
land_data_path = r'E:\land'
land_types = ['cropland', 'wetland', 'grass', 'forest', 'shrub']
years = [1990, 2000, 2010, 2019]
output_file = "land_use_sankey_diagram_multi_period.png"
pixel_area = 1540.13  # 每个像素实际面积

print("开始加载土地利用数据...")
data = load_land_use_data(land_data_path, years, land_types)

transition_matrices = []
for i in range(len(years) - 1):
    start_year, end_year = years[i], years[i+1]
    print(f"计算 {start_year}-{end_year} 转移矩阵...")
    transition_matrix = calculate_transition_matrix(data[start_year], data[end_year], land_types,
pixel_area)
    transition_matrices.append(transition_matrix)
```

```
    print_transition_matrix(transition_matrix, land_types)

print("创建多时间段 Sankey 图...")
create_sankey_diagram(transition_matrices, years, land_types, output_file)

print(f"处理完成。您可以在 {output_file} 查看结果。")
```

问题 2：

1. 数据清理（此步提取逐日数据，为问题 2 解答做准备）

```
import os
import numpy as np
import pandas as pd
import rasterio
from rasterio.warp import reproject, Resampling
import xarray as xr

elevation_path = r"E:\TIFF\chinadem_geo.tif"  # 高程数据
precipitation_path = r"E:\CHM_PRE_0.25dg_19612022.nc"  # 降水 NetCDF 数据
temperature_folder = r"E:\日平均数据\2018_avg"  # 2018 年的气温数据文件夹
output_folder = r"E:\csv"  # 输出的CSV 文件路径

# 重采样
with rasterio.open(elevation_path) as src:
    elevation = src.read(1).astype('float32')
    elevation_transform = src.transform
    elevation[elevation == src.nodata] = np.nan

    # 定义0.25 度分辨率
    target_resolution = 0.25
    target_transform = rasterio.transform.from_bounds(
        src.bounds.left, src.bounds.bottom, src.bounds.right, src.bounds.top,
        width=256, height=144  # 根据你提供的图像中的维度
    )

    elevation_resampled = np.empty((144, 256), dtype=np.float32)
    reproject(
        source=elevation,
        destination=elevation_resampled,
        src_transform=elevation_transform,
        src_crs=src.crs,
        dst_transform=target_transform,
        dst_crs=src.crs,
        resampling=Resampling.bilinear  # 双线性插值
    )

# 降水
precipitation_data = xr.open_dataset(precipitation_path)
precip_2018 = precipitation_data['pre'].sel(time=slice('2018-01-01', '2018-12-
31')).mean(dim='time')
precip_2018 = precip_2018.where(precip_2018 != -99.9, 0) # 将无效值 -99.9 替换为 0


#气温
```

```
tif_files = [os.path.join(temperature_folder, f) for f in os.listdir(temperature_folder) if
f.endswith('.tif')]
yearly_temperature = []

for tif_file in tif_files:
    with rasterio.open(tif_file) as src_temp:
        temp_data = src_temp.read(1).astype('float32')
        temp_data[temp_data == src_temp.nodata] = np.nan

        temp_resampled = np.empty((144, 256), dtype=np.float32)
        reproject(
            source=temp_data,
            destination=temp_resampled,
            src_transform=src_temp.transform,
            src_crs=src_temp.crs,
            dst_transform=target_transform,
            dst_crs=src_temp.crs,
            resampling=Resampling.bilinear  # 重采样到0.25 度
        )
        yearly_temperature.append(temp_resampled)

temperature_2018 = np.nanmean(yearly_temperature, axis=0)

# 获取坐标
longitudes, latitudes = np.meshgrid(
    np.linspace(target_transform.c, target_transform.c + target_resolution * 255, 256),
    np.linspace(target_transform.f, target_transform.f + target_resolution * 143, 144)
)

# 高程
elevation_df = pd.DataFrame({
    'Longitude': longitudes.ravel(),
    'Latitude': latitudes.ravel(),
    'Elevation': elevation_resampled.ravel()
})
elevation_df.to_csv(os.path.join(output_folder, 'elevation_0.25deg_2018.csv'), index=False)

precipitation_df = pd.DataFrame({
    'Longitude': longitudes.ravel(),
    'Latitude': latitudes.ravel(),
    'Precipitation': precip_2018.values.ravel()
})
precipitation_df.to_csv('precipitation_0.25deg_20181.csv', index=False)

temperature_df = pd.DataFrame({
    'Longitude': longitudes.ravel(),
    'Latitude': latitudes.ravel(),
    'Temperature': temperature_2018.ravel()
})
temperature_df.to_csv('temperature_0.25deg_2018.csv', index=False)

print("CSV 文件已经成功生成。")
```

2. 构建 gumbel 分布

```
import xarray as xr
import numpy as np
import pandas as pd
import rasterio
from rasterio.warp import reproject, Resampling
from scipy.stats import gumbel_r
import matplotlib.pyplot as plt

precipitation_path = r"E:\CHM_PRE_0.25dg_19612022.nc"
elevation_path = r"E:\TIFF\chinadem_geo.tif"
output_folder = r"E:\csv\2010"  # 输出的CSV 文件路径

plt.rcParams['font.sans-serif'] = ['SimHei']
plt.rcParams['axes.unicode_minus'] = False

with rasterio.open(elevation_path) as src:
    elevation = src.read(1).astype('float32')
    elevation_transform = src.transform
    elevation[elevation == src.nodata] = np.nan

    # 定义0.25 度分辨率
    target_resolution = 0.25
    target_transform = rasterio.transform.from_bounds(
        src.bounds.left, src.bounds.bottom, src.bounds.right, src.bounds.top,
        width=256, height=144  # 根据你提供的图像中的维度
    )

    elevation_resampled = np.empty((144, 256), dtype=np.float32)
    reproject(
        source=elevation,
        destination=elevation_resampled,
        src_transform=elevation_transform,
        src_crs=src.crs,
        dst_transform=target_transform,
        dst_crs=src.crs,
        resampling=Resampling.bilinear  # 双线性插值
    )

longitudes, latitudes = np.meshgrid(
    np.linspace(target_transform.c, target_transform.c + target_resolution * 255, 256),
    np.linspace(target_transform.f, target_transform.f + target_resolution * 143, 144)
)

precipitation_data = xr.open_dataset(precipitation_path)

# 选择2018 年的降水数据
precip = precipitation_data['pre'].sel(time=slice('1990-01-01', '2015-12-31'))

# 将无效值-99.9 替换为0
precip = precip.where(precip != -99.9, 0)

# 拟合Gumbel 分布并计算阈值
annual_max = precip.max(dim='time')  # 获取每个格点的年最大降水量
params = gumbel_r.fit(annual_max.values.ravel())  # 拟合Gumbel 分布参数
```

```
threshold = gumbel_r.ppf(0.95, *params)  # 计算95%分位数作为阈值

# 使用Gumbel 阈值判断极端降水事件
extreme_precip_events = (precip > threshold).astype(int)

extreme_event_counts = extreme_precip_events.sum(dim='time')

extreme_precip_resampled = np.empty((144, 256), dtype=np.float32)
reproject(
    source=extreme_event_counts.values,
    destination=extreme_precip_resampled,
    src_transform=precipitation_data['pre'].rio.transform(),
    src_crs="EPSG:4326",  # 指定源坐标参考系统
    dst_transform=target_transform,
    dst_crs="EPSG:4326",  # 指定目标坐标参考系统
    resampling=Resampling.nearest  # 使用nearest 避免数据被平滑
)
extreme_precip_df = pd.DataFrame({
    'Longitude': longitudes.ravel(),
    'Latitude': latitudes.ravel(),
    'Extreme_Rain_Events': extreme_precip_resampled.ravel()
})

extreme_precip_df.to_csv(f"{output_folder}/extreme_rain_events_20101.csv", index=False)

print("极端降水事件数据已成功导出为 CSV 文件。")

# 绘制Gumbel CDF
x = np.linspace(min(annual_max), max(annual_max), 1000)
cdf = gumbel_r.cdf(x, *params)
plt.figure(figsize=(8, 5))
plt.plot(x, cdf, label='Gumbel 累积分布函数')
plt.scatter(annual_max, gumbel_r.cdf(annual_max, *params), color='red', s=10, label='数据点')
plt.title('Gumbel 累积分布函数图示')
plt.xlabel('年最大降水量 (mm)')
plt.ylabel('累积概率')
plt.legend()
plt.grid(True)
plt.savefig('E:\gumbel')
plt.show()
```

3. 多元线性回归拟合

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from sklearn.svm import SVR
from sklearn.model_selection import GridSearchCV
import matplotlib.pyplot as plt
from sklearn.tree import plot_tree
```

```
plt.rcParams['font.sans-serif'] = ['SimHei']
plt.rcParams['axes.unicode_minus'] = False

def load_and_preprocess_data(file_path):
    data = pd.read_excel(file_path)
    data = data[(data[['Elevation', 'Precip', 'Temperature', 'Extreme_Rain_Events']] != 0).all(axis=1)]
    X = data[['Elevation', 'Precip', 'Temperature']]
    Y = data['Extreme_Rain_Events']
    return X, Y

def create_poly_features(X):
    poly = PolynomialFeatures(include_bias=False)
    X_poly = poly.fit_transform(X)
    return X_poly, poly

def split_data(X, Y, test_size=0.2, random_state=42):
    return train_test_split(X, Y, test_size=test_size, random_state=random_state)

————————————————多元线性，随机森林与支持向量机模型比较————————————————
def linear_regression(X_train, X_test, Y_train, Y_test, poly, original_feature_names):
    model = LinearRegression()
    model.fit(X_train, Y_train)
    Y_pred = model.predict(X_test)
    r2 = r2_score(Y_test, Y_pred)
    print(f"R-squared with polynomial features: {r2}")

    feature_names = poly.get_feature_names_out(original_feature_names)
    coefficients = model.coef_
    intercept = model.intercept_

    equation = f"Y = {intercept:.4f}"
    for name, coef in zip(feature_names, coefficients):
        if coef >= 0:
            equation += f" + {coef:.4f}*{name}"
        else:
            equation += f" - {abs(coef):.4f}*{name}"

    print("Multi-variable Linear Regression Equation:")
    print(equation)

    return model, Y_pred

def random_forest(X_train, X_test, Y_train, Y_test):
    param_grid = {
        'n_estimators': [300],
        'max_depth': [20],
        'min_samples_split': [ 20]
    }
    rf = RandomForestRegressor(random_state=42)
    grid_search = GridSearchCV(estimator=rf, param_grid=param_grid, cv=5, scoring='r2',
verbose=2, n_jobs=-1)
    grid_search.fit(X_train, Y_train)

    best_rf = grid_search.best_estimator_
    rf_pred = best_rf.predict(X_test)
    rf_r2 = r2_score(Y_test, rf_pred)
    print(f"Best parameters: {grid_search.best_params_}")
```

```
    print(f"R-squared (Random Forest): {rf_r2}")
    return best_rf, rf_pred


# Train and evaluate SVR model
def svr_model(X_train, X_test, Y_train, Y_test):
    svr = SVR(kernel='rbf')
    svr.fit(X_train, Y_train)
    svr_pred = svr.predict(X_test)
    svr_r2 = r2_score(Y_test, svr_pred)
    print(f"R-squared (Support Vector Regression): {svr_r2}")
    return svr, svr_pred


# Visualize decision tree
def visualize_tree(rf_model, feature_names, save_path):
    plt.figure(figsize=(60, 18))
    plot_tree(rf_model.estimators_[0],
              feature_names=feature_names,
              filled=True,
              rounded=True,
              max_depth=3,
              fontsize=25)
    plt.title('Visualizing Decision Tree in Random Forest')
    plt.savefig(save_path)
    plt.show()
    plt.close()

def plot_feature_importances(rf_model, feature_names, save_path):
    feature_importances = pd.Series(rf_model.feature_importances_, index=feature_names)
    feature_importances.nlargest(10).sort_values().plot(kind='barh')
    plt.title('Top 10 Feature Importances in Random Forest Model')
    plt.xlabel('Importance')
    plt.savefig(save_path)
    plt.close()

def plot_actual_vs_predicted(Y_test, linear_pred, rf_pred, svr_pred, save_path):
    fig, ax = plt.subplots(1, 3, figsize=(25, 6))
    models = [('Linear Regression', linear_pred, 'blue'),
              ('Random Forest', rf_pred, 'red'),
              ('Support Vector Regression', svr_pred, 'green')]

    for i, (name, pred, color) in enumerate(models):
        ax[i].scatter(Y_test, pred, color=color)
        ax[i].plot([Y_test.min(), Y_test.max()], [Y_test.min(), Y_test.max()], 'k--', lw=4)
        ax[i].set_title(f'Actual vs. Predicted ({name})')
        ax[i].set_xlabel('Actual')
        ax[i].set_ylabel('Predicted')

    plt.savefig(save_path)
    plt.close()


def main():
    X, Y = load_and_preprocess_data('results_2009_2010_with_disaster_level.xlsx')
    X_poly, poly = create_poly_features(X)
    X_train, X_test, Y_train, Y_test = split_data(X_poly, Y)
```

```
    linear_model, linear_pred = linear_regression(X_train, X_test, Y_train, Y_test, poly, X.columns)
    rf_model, rf_pred = random_forest(X_train, X_test, Y_train, Y_test)
    svr, svr_pred = svr_model(X_train, X_test, Y_train, Y_test)

    visualize_tree(rf_model, poly.get_feature_names_out(X.columns),
'E:/Visualizing_Decision_Tree_in_Random_Forest.png')
    plot_feature_importances(rf_model, poly.get_feature_names_out(X.columns),
'E:/Feature_Importances.png')
    plot_actual_vs_predicted(Y_test, linear_pred, rf_pred, svr_pred, 'E:/Actual_vs_Predicted.png')

if __name__ == "__main__":
    main()


————————————————————————二元线性回归————————————————————————
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score
import numpy as np
from sklearn.metrics import mean_squared_error
import statsmodels.api as sm
import matplotlib.pyplot as plt

plt.rcParams['font.sans-serif'] = ['SimHei']
plt.rcParams['axes.unicode_minus'] = False

elevation_df = pd.read_csv(r'E:\csv\2010\elevation_0.25deg_2010.csv')
precipitation_df = pd.read_csv(r'E:\csv\2010\precipitation_0.25deg_2010.csv')
temperature_df = pd.read_csv(r'E:\csv\2010\temperature_0.25deg_2010.csv')
#heavy_rain_events_df = pd.read_csv(r'E:\2010\test\heavy_rain_events_2010.csv')

data = pd.DataFrame({
    'Longitude': elevation_df['Longitude'],
    'Latitude': elevation_df['Latitude'],
    'Elevation': elevation_df['Elevation'],
    'Precipitation': precipitation_df['Precipitation'],
    'Temperature': temperature_df['Temperature'],
    #'HeavyRainEvents': heavy_rain_events_df['Heavy_Rain_Events']
})

# 删除任何在高程、降水量、温度或重大降雨事件为0 的行
data = data[(data[['Elevation', 'Precipitation', 'Temperature']] != 0).all(axis=1)]
data.dropna(inplace=True)

X = data[['Temperature', 'Elevation']]
Y = data['Precipitation']

X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=42)

model = LinearRegression()
model.fit(X_train, Y_train)

Y_pred = model.predict(X_test)

r2 = r2_score(Y_test, Y_pred)
```

```
print(f"R 方值: {r2}")
print("系数:", model.coef_)
print("截距:", model.intercept_)
coefficients = model.coef_
intercept = model.intercept_

regression_formula = f"P = {intercept:.3f} + ({coefficients[0]:.3f}) * T + ({coefficients[1]:.3f}) * H"
print("回归方程:")
print(regression_formula)

Y_pred = model.predict(X_test)
r2 = r2_score(Y_test, Y_pred)
mse = mean_squared_error(Y_test, Y_pred)
print(f"R 方值: {r2}")
print(f"均方误差: {mse}")

residuals = Y_test - Y_pred
plt.figure(figsize=(10, 6))
plt.scatter(Y_pred, residuals, color='blue')
plt.axhline(y=0, color='red', linestyle='--')
plt.xlabel('预测的降水量')
plt.ylabel('残差')
plt.title('残差图')
plt.savefig('E:\残差图')
plt.show()

X_train_sm = sm.add_constant(X_train)  # 添加截距项
model_sm = sm.OLS(Y_train, X_train_sm).fit()
print(model_sm.summary())

fig, ax = plt.subplots(figsize=(6, 4))
sm.qqplot(model_sm.resid, line='s', ax=ax)  # Q-Q 图
ax.set_title('残差的 Q-Q 图')
plt.savefig('E:\残差的 Q-Q 图')
plt.show()

fig = plt.figure(figsize=(12, 8))
ax = fig.add_subplot(111, projection='3d')
sc = ax.scatter(X_test['Temperature'], X_test['Elevation'], Y_test, c='blue', label='实际',
depthshade=False)
sc = ax.scatter(X_test['Temperature'], X_test['Elevation'], Y_pred, c='red', label='预测',
depthshade=False)
temp_range = np.linspace(X_test['Temperature'].min(), X_test['Temperature'].max(), 20)
elev_range = np.linspace(X_test['Elevation'].min(), X_test['Elevation'].max(), 20)
temp_grid, elev_grid = np.meshgrid(temp_range, elev_range)
Z = model.predict(np.array([temp_grid.ravel(), elev_grid.ravel()]).T).reshape(temp_grid.shape)
ax.plot_surface(temp_grid, elev_grid, Z, color='gray', alpha=0.5)

ax.set_xlabel('温度 (°C)')
ax.set_ylabel('高程 (m)')
ax.set_zlabel('降水量 (mm)')
plt.title('2010 年温度、海拔与降水量的三维回归分析')
plt.legend()
```

```
plt.savefig('E:\\2010 年温度、海拔与降水量的三维回归分析.png')
plt.show()
```

问题 3

数据整理（多年逐日数据过大，整理出多年格点均值，便于预测）（该文件重提取栅格的代码改编自 https://blog.csdn.net/mm15707472721/article/details/135513655

1. ）

```
import xarray as xr
import numpy as np
import rasterio
from rasterio.warp import reproject, Resampling
import os
import pandas as pd
import matplotlib.pyplot as plt

import csv

plt.rcParams['font.sans-serif'] = ['SimHei']

# 避免负号显示为方块
plt.rcParams['axes.unicode_minus'] = False

def resample_raster(src, dst_crs, dst_shape, dst_transform):
    """重采样栅格数据"""
    resampled = np.zeros(dst_shape)
    reproject(
        source=src.read(1),
        destination=resampled,
        src_transform=src.transform,
        src_crs=src.crs,
        dst_transform=dst_transform,
        dst_crs=dst_crs,
        resampling=Resampling.bilinear
    )
    return resampled


def generate_heatmaps(data, variables, year, num_cols=3):
    year_data = data[data['年'] == year]
    num_rows = (len(variables) + num_cols - 1) // num_cols
    fig, axes = plt.subplots(num_rows, num_cols, figsize=(6 * num_cols, 5 * num_rows))
    fig.suptitle(f'变量热力图 - {year}年', fontsize=16)
    axes = axes.flatten() if num_rows > 1 or num_cols > 1 else [axes]

    for i, var in enumerate(variables):
        pivot = year_data.pivot(index='纬度', columns='经度', values=var)
        im = axes[i].imshow(pivot.iloc[::-1], cmap='viridis', aspect='auto', origin='lower')
        axes[i].set_title(var)
        plt.colorbar(im, ax=axes[i])

    for j in range(i + 1, len(axes)):
        axes[j].axis('off')

    plt.tight_layout()
    plt.savefig(r'E:\alldata')
```

```
    plt.show()

def load_and_process_data(start_year, end_year, output_path):
    precipitation_path = r"E:\CHM_PRE_0.25dg_19612022.nc"
    elevation_path = r"E:\TIFF\chinadem_geo.tif"
    temperature_folder = r"E:\日平均数据"
    land_data_path = r'E:\land'
    pop_data_path = r'E:\pop_output'
    gdp_data_path = r'E:\GDP_output'

    precipitation_data = xr.open_dataset(precipitation_path)
    precip = precipitation_data['pre'].sel(time=slice(f'{start_year}-01-01', f'{end_year}-12-31'))

    lon_min, lon_max = precip.longitude.min().item(), precip.longitude.max().item()
    lat_min, lat_max = precip.latitude.min().item(), precip.latitude.max().item()

    dst_crs = 'EPSG:4326'  # WGS84
    dst_res = 0.25  # 0.25 度分辨率
    dst_width = precip.longitude.size
    dst_height = precip.latitude.size
     dst_transform = rasterio.transform.from_bounds(lon_min, lat_min, lon_max, lat_max, dst_width,
dst_height)

    print(f"Grid size: {dst_height} x {dst_width}")

    # 读取高程数据并重采样
    with rasterio.open(elevation_path) as src_elev:
        elevation = resample_raster(src_elev, dst_crs, (dst_height, dst_width), dst_transform)

    years = range(start_year, end_year + 1)

    # 处理气温数据
    temp_data = []
    for year in years:
        year_folder = os.path.join(temperature_folder, f'{year}_avg')
        if not os.path.exists(year_folder):
            print(f"No temperature data available for year {year}.")
            temp_data.append(np.full((dst_height, dst_width), np.nan))
            continue

        tif_files = [os.path.join(year_folder, f) for f in os.listdir(year_folder) if f.endswith('.tif')]
        yearly_temperature = []

        for tif_file in tif_files:
            with rasterio.open(tif_file) as src_temp:
                temp = resample_raster(src_temp, dst_crs, (dst_height, dst_width), dst_transform)
                yearly_temperature.append(temp)

        temp_data.append(np.mean(yearly_temperature, axis=0))

    # 处理土地利用数据
    land_types = ['cropland', 'wetland', 'grass', 'forest', 'shrub']
    land_use_data = {}

    for land_type in land_types:
        land_type_data = []
```

```
    for year in years:
        file_name = f"{land_type}-{year}.tif"
        file_path = os.path.join(land_data_path, file_name)

        if os.path.exists(file_path):
            with rasterio.open(file_path) as src:
                data = resample_raster(src, dst_crs, (dst_height, dst_width), dst_transform)
                land_type_data.append(data)
        else:
            print(f"No land use data available for {land_type} in year {year}.")
            land_type_data.append(np.full((dst_height, dst_width), np.nan))

    land_use_data[land_type] = np.array(land_type_data)

# 处理GDP 数据
gdp_data = []
for year in years:
    file_name = f"gdp{year}.tif"
    file_path = os.path.join(gdp_data_path, file_name)
    if os.path.exists(file_path):
        with rasterio.open(file_path) as src:
            resampled_data = resample_raster(src, dst_crs, (dst_height, dst_width), dst_transform)
            gdp_data.append(resampled_data)
    else:
        print(f"GDP data file not found for year {year}")
        gdp_data.append(np.full((dst_height, dst_width), np.nan))

# 处理人口数据
pop_data = []
for year in years:
    folder_name = f"pop{year}"
    file_path = os.path.join(pop_data_path, folder_name, f'{folder_name}.tif')
    if os.path.exists(file_path):
        with rasterio.open(file_path) as src:
            resampled_data = resample_raster(src, dst_crs, (dst_height, dst_width), dst_transform)
            pop_data.append(resampled_data)
    else:
        print(f"Population data file not found for year {year}")
        pop_data.append(np.full((dst_height, dst_width), np.nan))

# 创建经纬度网格
lon, lat = np.meshgrid(precip.longitude, precip.latitude)

# 创建CSV 文件并写入表头
with open(output_path, 'w', newline='') as csvfile:
    writer = csv.writer(csvfile)
    writer.writerow(
        ['年', '经度', '纬度', '人口', 'GDP', '耕地', '森林', '草地', '灌木', '湿地', '降水', '温度', '高程',
         '是否中国', '土地利用系数'])

# 按年份处理数据
for i, year in enumerate(years):
    print(f"Processing year {year}...")

    year_precip = precip.isel(time=precip.time.dt.year == year).mean(dim='time').values
    year_precip = np.flipud(year_precip)
```

```
        year_precip[year_precip < -98] = 0


        year_data = pd.DataFrame({
            '年': year,
            '经度': lon.ravel(),
            '纬度': lat.ravel(),
            '人口': pop_data[i].ravel(),
            'GDP': gdp_data[i].ravel(),
            '耕地': land_use_data['cropland'][i].ravel(),
            '森林': land_use_data['forest'][i].ravel(),
            '草地': land_use_data['grass'][i].ravel(),
            '灌木': land_use_data['shrub'][i].ravel(),
            '湿地': land_use_data['wetland'][i].ravel(),
            '降水': year_precip.ravel(),
            '温度': temp_data[i].ravel(),
            '高程': elevation.ravel()
        })


        year_data['在中国'] = 1
        year_data.loc[(year_data.人口 < 0) | (year_data.GDP < 0) |
                      ((year_data.耕地 == 0) & (year_data.森林 == 0) & (year_data.草地 == 0) &
                       (year_data.灌木 == 0) & (year_data.湿地 == 0)), '在中国'] = 0

————————————————————计算土地利用系数————————————————————

        year_data['土地利用系数'] = 0.28 * year_data['耕地'] + 0.09 * year_data['森林'] + 0.2 *
year_data['草地'] + \
                                0.24 * year_data['灌木'] + 0.19 * year_data['湿地']

        # 筛选中国陆地数据并排序
        year_data = year_data[year_data.在中国 == 1].sort_values(by=['经度', '纬度'])

        # 将数据追加到CSV文件
        year_data.to_csv(output_path, mode='a', header=False, index=False)

    print(f"Data processing complete. Results saved to CSV file: {output_path}")

    return output_path


if __name__ == "__main__":
    output_file = load_and_process_data(1990, 2015, output_path='../q4/results_1990_2015.csv')
    print(f"Data saved to {output_file}")
# print(data.head())
# print(data.describe())
```

2. logistic 回归

```
import pandas as pd
from sklearn.linear_model import LogisticRegression
from sklearn.impute import SimpleImputer
from sklearn.pipeline import Pipeline
import matplotlib.pyplot as plt
import cartopy.crs as ccrs
import cartopy.feature as cfeature

plt.rcParams['font.sans-serif'] = ['SimHei']
plt.rcParams['axes.unicode_minus'] = False

def create_extreme_weather_label(data, precipitation_threshold=0.95):
    threshold = data['降水'].quantile(precipitation_threshold)
    data['极端天气'] = (data['降水'] > threshold).astype(int)
    return data


def train_model_and_predict(data, features, target, threshold=0.8):
    X = data[features]
    y = data[target]

    pipeline = Pipeline([
        ('imputer', SimpleImputer(strategy='mean')),  # 使用平均值填充缺失值
        ('classifier', LogisticRegression(random_state=42, max_iter=1000))
    ])

    pipeline.fit(X, y)
    logistic_model = pipeline.named_steps['classifier']

    print("\n 逻辑回归模型参数:")
    for feature, coef in zip(features, logistic_model.coef_[0]):
        print(f"{feature}: {coef:.4f}")

    print(f"截距: {logistic_model.intercept_[0]:.4f}")

    equation = "log(p/(1-p)) = "
    for feature, coef in zip(features, logistic_model.coef_[0]):
        equation += f"{coef:.4f} * {feature} + "
    equation += f"{logistic_model.intercept_[0]:.4f}"
    print("\n 逻辑回归方程:")
    print(equation)

    pred_proba = pipeline.predict_proba(X)
    data['pred'] = pred_proba[:, 1]
    data['is_extreme_weather'] = (data['pred'] > threshold).astype(int)

    return data


def visualize_extreme_weather(data, year):
    plt.figure(figsize=(12, 9))
    ax = plt.axes(projection=ccrs.PlateCarree())
```

```
    ax.set_extent([73, 130, 15, 54], crs=ccrs.PlateCarree())
    ax.add_feature(cfeature.LAND)
    ax.add_feature(cfeature.OCEAN)
    ax.add_feature(cfeature.COASTLINE)
    ax.add_feature(cfeature.LAKES, alpha=0.5)
    ax.add_feature(cfeature.RIVERS)

    # 绘制极端天气地点
    extreme_weather_locations = data[data['is_extreme_weather'] == 1]

    # 计算纬度的最大值和最小值
    lat_min, lat_max = data['纬度'].min(), data['纬度'].max()

    # 反转纬度
    reversed_latitudes = lat_max + lat_min - extreme_weather_locations['纬度']

    plt.scatter(extreme_weather_locations['经度'], reversed_latitudes,
                color='red', s=10, alpha=0.7, transform=ccrs.PlateCarree())

    plt.title(f"{year}年预测的暴雨灾害地点")
    plt.show()


def process_data_and_visualize(excel_path, year):
    data = pd.read_excel(excel_path)

    data = create_extreme_weather_label(data)
    features = ['高程', '降水', '土地利用系数', '经度', '纬度']
    target = '极端天气'
    result = train_model_and_predict(data, features, target)

    # 筛选指定年份的数据进行可视化
    year_data = result[result['年'] == year]
    visualize_extreme_weather(year_data, year)

    return result

if __name__ == "__main__":
    excel_path = r'E:\future_predictions_2025_2035.csv'
    year = 2010

    result = process_data_and_visualize(excel_path, year)
    print(f"预测的极端天气事件总数: {result['is_extreme_weather'].sum()}")
    print(f"{year}年预测的极端天气事件数量: {result[result['年'] == year]
['is_extreme_weather'].sum()}")

    # 输出每个特征的缺失值数量
    features = ['高程', '降水', '土地利用系数', '经度', '纬度']
    for feature in features:
        missing = result[feature].isnull().sum()
        print(f"{feature} 的缺失值数量: {missing}")
```

3. ARIMA+线性回归预测

```
import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt
from tqdm import tqdm
import os


def predict_future_land_use(data, past_years, future_years):
    """使用简单线性回归预测未来土地利用系数"""
    X = np.array(range(past_years)).reshape(-1, 1)
    y = data['土地利用系数'].values

    model = LinearRegression()
    model.fit(X, y)

    future_X = np.array(range(past_years, past_years + future_years)).reshape(-1, 1)
    future_y = model.predict(future_X)

    return future_y


def predict_future_precipitation(data, ratios):
    """根据给定的比例预测未来降水，使用2015 年作为基准"""
    base_year_data = data[data['年'] == 2015]
    if base_year_data.empty:
        base_year = data['年'].max()
        base_year_data = data[data['年'] == base_year]

    if base_year_data.empty:
        raise ValueError(f"No valid base year data found for location")

    base_year_precipitation = base_year_data['降水'].values[0]
    future_precipitation = [base_year_precipitation * ratio for ratio in ratios]
    return future_precipitation


def process_grid_data(data, past_years=10, future_years=11, precipitation_ratios=None):
    """对每个格点进行分析和预测"""
    if precipitation_ratios is None or len(precipitation_ratios) != future_years:
        raise ValueError("precipitation_ratios must be provided and match the number of future
years")

    unique_locations = data.groupby(['经度', '纬度'])

    future_data = []
    total_locations = len(unique_locations)

    for (lon, lat), group in tqdm(unique_locations, total=total_locations, desc="Processing
locations"):
        if len(group) < past_years:
            print(f"Skipping location (经度 {lon}, 纬度 {lat}) due to insufficient data")
            continue
```

```
        try:
            land_use_predictions = predict_future_land_use(group.iloc[-past_years:], past_years,
future_years)
            precipitation_predictions = predict_future_precipitation(group, precipitation_ratios)
            elevation = group['高程'].iloc[0]  # 获取该格点的高程数据

            for year, land_use_pred, precip_pred in zip(range(2025, 2025 + future_years),
land_use_predictions,
                                                        precipitation_predictions):
                future_data.append({
                    '年': year,
                    '经度': lon,
                    '纬度': lat,
                    '高程': elevation,  # 添加高程数据
                    '降水': precip_pred,
                    '土地利用系数': land_use_pred
                })
        except Exception as e:
            print(f"Error processing location (经度 {lon}, 纬度 {lat}): {str(e)}")

    return pd.DataFrame(future_data)


if __name__ == "__main__":
    try:
        # 读取数据，添加'高程'列
        data = pd.read_csv('../q4/results_2005_2015.csv', encoding='gbk',
                           usecols=['年', '经度', '纬度', '高程', '降水', '土地利用系数'])
        print(f"Loaded data shape: {data.shape}")
        print(f"Columns: {data.columns}")

        # 设置降水预测的比例（2025-2035 年，共11 年）
        precipitation_ratios = [1.090079612, 1.092903732, 1.09572621, 1.09855033, 1.101374451,
1.104198571, 1.107020979,
                                1.1098451, 1.11266922, 1.115493341, 1.118317461]

        # 进行分析和预测
        future_data = process_grid_data(data, past_years=10, future_years=11,
precipitation_ratios=precipitation_ratios)
        output_path = r'E:\future_predictions_2025_2035.csv'
        future_data.to_csv(output_path, index=False, encoding='gbk')

        print(f"分析完成，结果已保存到 '{output_path}'")

        # 可视化结果（2035 年的降水和土地利用系数）
        for feature in ['降水', '土地利用系数']:
            plt.figure(figsize=(12, 8))
            year_2035 = future_data[(future_data['年'] == 2035) & (future_data[feature].notna())]
            plt.scatter(year_2035['经度'], year_2035['纬度'], c=year_2035[feature], cmap='YlOrRd',
s=10)
            plt.colorbar(label=feature)
            plt.title(f'2035 年{feature}预测')
            plt.xlabel('经度')
```

```
        plt.ylabel('纬度')
        plt.savefig(os.path.join('E:', f'2035_{feature}_prediction.png'))
        plt.show()
    except Exception as e:
        print(f"An error occurred: {str(e)}")
        import traceback

        traceback.print_exc()
```

4. 随机森林预测

```
import pandas as pd
import numpy as np
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import TimeSeriesSplit
from sklearn.metrics import r2_score, mean_squared_error
from sklearn.preprocessing import StandardScaler
import matplotlib.pyplot as plt
from tqdm import tqdm
import os

plt.rcParams['font.sans-serif'] = ['SimHei']
plt.rcParams['axes.unicode_minus'] = False


def load_and_preprocess_data(file_path):
    data = pd.read_csv(file_path, encoding='gbk', usecols=['年', '经度', '纬度', '高程', '降水', '土地利用系数'])
    data['年'] = data['年'].astype(int)
    data = data.sort_values(['经度', '纬度', '年'])
    data = data.dropna()
    data = data.drop_duplicates()
    print(f"数据范围：从 {data['年'].min()} 年到 {data['年'].max()} 年")
    print(f"总数据点数：{len(data)}")
    print(f"唯一位置数：{data.groupby(['经度', '纬度']).ngroups}")
    return data


def create_features(group, feature):
    group['年_sin'] = np.sin(2 * np.pi * group['年'] / group['年'].max())
    group['年_cos'] = np.cos(2 * np.pi * group['年'] / group['年'].max())
    group[f'{feature}_diff'] = group[feature].diff()
    group[f'{feature}_lag1'] = group[feature].shift(1)
    group[f'{feature}_rolling_mean'] = group[feature].rolling(window=3, min_periods=1).mean()
    return group.dropna()


def process_grid_data(data, future_years):
    unique_locations = data.groupby(['经度', '纬度'])
    future_data_rf = []
    all_r2_rf = {'土地利用系数': [], '降水': []}
    all_mse_rf = {'土地利用系数': [], '降水': []}

    rf_model = RandomForestRegressor(n_estimators=100, max_depth=None, min_samples_leaf=2,
```

```
random_state=42)

    for (lon, lat), group in tqdm(unique_locations, desc="Processing locations"):
        if len(group) < 5:
            continue

        elevation = group['高程'].iloc[0]

        for feature in ['土地利用系数', '降水']:
            feature_group = create_features(group.copy(), feature)

            X = feature_group[
                ['年', '年_sin', '年_cos', f'{feature}_diff', f'{feature}_lag1',
f'{feature}_rolling_mean']].values
            y = feature_group[feature].values

            if len(np.unique(y)) < 2:
                continue

            scaler = StandardScaler()
            X_scaled = scaler.fit_transform(X)

            tscv = TimeSeriesSplit(n_splits=3)
            cv_scores = []

            for train_index, test_index in tscv.split(X_scaled):
                X_train, X_test = X_scaled[train_index], X_scaled[test_index]
                y_train, y_test = y[train_index], y[test_index]

                rf_model.fit(X_train, y_train)
                rf_pred = rf_model.predict(X_test)
                cv_scores.append(r2_score(y_test, rf_pred))

            rf_r2 = np.mean(cv_scores)
            rf_mse = mean_squared_error(y_test, rf_pred)

            all_r2_rf[feature].append(rf_r2)
            all_mse_rf[feature].append(rf_mse)

            # 预测未来值
            last_year = group['年'].max()
            future_years_range = range(last_year + 1, last_year + 1 + future_years)
            future_X = []
            for year in future_years_range:
                year_sin = np.sin(2 * np.pi * year / group['年'].max())
                year_cos = np.cos(2 * np.pi * year / group['年'].max())
                if len(future_X) == 0:
                    diff = 0
                    lag1 = y[-1]
                    rolling_mean = y[-1]
                else:
                    diff = future_X[-1][3] - lag1
                    lag1 = future_X[-1][3]
                    rolling_mean = np.mean([x[3] for x in future_X[-3:]] + [y[-1]])
                future_X.append([year, year_sin, year_cos, diff, lag1, rolling_mean])
```

```
        future_X_scaled = scaler.transform(future_X)
        rf_future_y = rf_model.predict(future_X_scaled)

        for year, rf_pred in zip(future_years_range, rf_future_y):
            future_data_rf.append({
                '年': year, '经度': lon, '纬度': lat, '高程': elevation, feature: rf_pred
            })


if __name__ == "__main__":
    file_path = '../q4/results_1990_2015.csv'
    data = load_and_preprocess_data(file_path)
    future_data_rf, all_r2_rf, all_mse_rf = process_grid_data(data, future_years=20)

    if not future_data_rf.empty:
        output_dir = r'E:\predictions'
        os.makedirs(output_dir, exist_ok=True)
        excel_file = os.path.join(output_dir, 'future_predictions_2025_2035_rf.xlsx')
        future_data_rf.to_excel(excel_file, index=False)
        print(f"分析完成，结果已保存到 '{excel_file}'")
    else:
        print("未能生成有效的预测。")

————————————————可视化格点时间序列————————————————————
def visualize_time_series(data, future_data, lon, lat, feature):
    plt.figure(figsize=(12, 6))
    plt.plot(data['年'], data[feature], 'bo-', label='历史数据')
    future_years = future_data['年']
    plt.plot(future_years, future_data[feature], 'ro--', label='预测数据')
    plt.title(f'{feature} 时间序列 (经度 {lon}, 纬度 {lat})')
    plt.xlabel('年份')
    plt.ylabel(feature)
    plt.legend()
    plt.grid(True)
    plt.show()
```

5. LSTM 模型（迫于硬件设施，不能在规定时间内完成预测）

```
import pandas as pd
import numpy as np
from sklearn.preprocessing import MinMaxScaler
from tensorflow import keras
import matplotlib.pyplot as plt
import traceback
import os
from tqdm import tqdm

def prepare_data(data, feature, sequence_length):
    X, y = [], []
    for i in range(len(data) - sequence_length):
        X.append(data[i:(i + sequence_length)])
        y.append(data[i + sequence_length])
    X = np.array(X).reshape(-1, sequence_length, 1)
    y = np.array(y)
    print(f"Feature: {feature}, X shape: {X.shape}, y shape: {y.shape}")
    return X, y
```

```
def create_lstm_model(sequence_length, n_features):
    model = keras.Sequential([
        keras.layers.LSTM(64, activation='relu', input_shape=(sequence_length, n_features),
return_sequences=True),
        keras.layers.LSTM(32, activation='relu'),
        keras.layers.Dense(16, activation='relu'),
        keras.layers.Dense(1)
    ])
    model.compile(optimizer=keras.optimizers.Adam(learning_rate=0.001), loss='mse')
    return model

def train_and_predict(data, feature, sequence_length, future_years):
    print(f"Training for feature: {feature}")
    print(f"Data shape: {data.shape}")

    if len(data) <= sequence_length:
        print(f"Insufficient data for feature {feature}. Skipping...")
        return None

    try:
        scaler = MinMaxScaler()
        scaled_data = scaler.fit_transform(data[[feature]])

        X, y = prepare_data(scaled_data, feature, sequence_length)

        if X.shape[0] == 0 or y.shape[0] == 0:
            print(f"No valid sequences created for feature {feature}. Skipping...")
            return None

        model = create_lstm_model(sequence_length, 1)
        history = model.fit(X, y, epochs=100, batch_size=32, verbose=0)

        print(f"Training completed. Final loss: {history.history['loss'][-1]}")

        last_sequence = scaled_data[-sequence_length:].reshape(1, sequence_length, 1)
        future_predictions = []

        for _ in range(future_years):
            next_pred = model.predict(last_sequence, verbose=0)
            future_predictions.append(next_pred[0, 0])
            last_sequence = np.roll(last_sequence, -1, axis=1)
            last_sequence[0, -1, 0] = next_pred[0, 0]

        return scaler.inverse_transform(np.array(future_predictions).reshape(-1, 1))
    except Exception as e:
        print(f"Error occurred while processing feature {feature}: {str(e)}")
        print("Traceback:")
        traceback.print_exc()
        return None

def process_grid_data(data, features, sequence_length=5, future_years=11):
    unique_locations = data.groupby(['经度', '纬度'])

    future_data = []
    total_locations = len(unique_locations)
```

```
    for i, ((lon, lat), group) in tqdm(enumerate(unique_locations), total=total_locations,
desc="Processing locations"):
        print(f"\nProcessing location {i + 1}/{total_locations}: 经度 {lon}, 纬度 {lat}")
        print(f"Group shape: {group.shape}")

        if len(group) <= sequence_length:
            print(f"Skipping location due to insufficient data (only {len(group)} points)")
            continue

        for feature in features:
            try:
                predictions = train_and_predict(group, feature, sequence_length, future_years)

                if predictions is not None:
                    for year, pred in zip(range(2025, 2025 + future_years), predictions):
                        future_data.append({
                            '年': year,
                            '经度': lon,
                            '纬度': lat,
                            feature: pred[0]
                        })
            except Exception as e:
                print(f"Error processing feature {feature} at location (经度 {lon}, 纬度 {lat}): {str(e)}")
                print("Traceback:")
                traceback.print_exc()

    return pd.DataFrame(future_data)

if __name__ == "__main__":
    try:
        data = pd.read_csv('../q4/results_1990_2015.csv', encoding='gbk',
                           usecols=['年', '经度', '纬度', '降水', '土地利用系数'])
        print(f"Loaded data shape: {data.shape}")
        print(f"Columns: {data.columns}")

        features = ['降水', '土地利用系数']
        future_data = process_grid_data(data, features, future_years=20)  # 预测11年（2025-2035）

        output_path = r'E:\future_predictions_2025_2035.xlsx'
        future_data.to_excel(output_path, index=False, engine='openpyxl')

        print(f"分析完成，结果已保存到 '{output_path}'")


————————————————可视化随机格点 LSTM 时间序列预测————————————————————————
def visualize_results(data, future_data, rmse_data, features):
    for feature in features:
        for location in future_data.groupby(['经度', '纬度']):
            lon, lat = location[0]
            location_data = location[1]

            # 获取历史数据
            historical_data = data[(data['经度'] == lon) & (data['纬度'] == lat)].sort_values('年')

            fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(12, 10), sharex=True)
```

```
        fig.suptitle(f'{feature} Forecast for Location ({lon}, {lat})', fontsize=16)

        # 绘制历史数据和预测数据
        ax1.plot(historical_data['年'], historical_data[feature], label='Observed', color='blue')
        ax1.plot(location_data['年'], location_data[feature], label='Forecast', color='red')
        ax1.set_ylabel(feature)
        ax1.legend()

        # 计算并绘制残差
        historical_years = historical_data['年'].tolist()
        forecast_years = location_data['年'].tolist()
        all_years = historical_years + forecast_years

        observed = historical_data[feature].tolist() + [np.nan] * len(forecast_years)
        predicted = [np.nan] * len(historical_years) + location_data[feature].tolist()

        residuals = np.array(observed) - np.array(predicted)

        ax2.stem(all_years, residuals, linefmt='cyan', markerfmt='co', bottom=0)
        ax2.axhline(y=0, color='r', linestyle='-')
        ax2.set_xlabel('Year')
        ax2.set_ylabel('Residuals')

        rmse = rmse_data.get((lon, lat, feature), {}).get('rmse', 'N/A')
        ax2.set_title(f'RMSE = {rmse:.4f}' if isinstance(rmse, float) else f'RMSE = {rmse}')

        plt.tight_layout()
        plt.savefig(os.path.join('E:', f'{feature}_forecast_{lon}_{lat}.png'))
        plt.close()

    # RMSE table
    rmse_df = pd.DataFrame([(lon, lat, feature, data['rmse'])
                            for (lon, lat, feature), data in rmse_data.items()],
                           columns=['经度', '纬度', 'Feature', 'RMSE'])
    rmse_df.to_csv(os.path.join('E:', 'rmse_results.csv'), index=False)
```

7.可视化预测数据

```
def visualize_heatmap(data, years, save_path_template):
    plt.rcParams['font.size'] = 16  # 增大全局字体大小

    for year in years:
        fig, ax = plt.subplots(figsize=(30, 20),
                               subplot_kw={'projection': ccrs.PlateCarree()})

        year_data = data[data['年'] == year]

        # 计算纬度的最大值和最小值
        lat_min, lat_max = year_data['纬度'].min(), year_data['纬度'].max()

        # 反转纬度
        reversed_latitudes = lat_max + lat_min - year_data['纬度']

        scatter = ax.scatter(year_data['经度'], reversed_latitudes,
                             c=year_data['暴雨成灾概率'], cmap='YlOrRd',
```

```
                    s=50, alpha=0.7,
                    transform=ccrs.PlateCarree(),
                    vmin=0, vmax=1)  # 设置颜色范围为0 到1

        # 设置标题
        ax.set_title(f'{year}年暴雨成灾概率分布', fontsize=36)

        # 添加经纬度网格线和标签
        gl = ax.gridlines(draw_labels=True, linewidth=0.5, color='gray', alpha=0.5, linestyle='--')
        gl.top_labels = False
        gl.right_labels = False
        gl.xformatter = LONGITUDE_FORMATTER
        gl.yformatter = LATITUDE_FORMATTER
        gl.xlabel_style = {'size': 24}
        gl.ylabel_style = {'size': 24}
        cbar = fig.colorbar(scatter, ax=ax, shrink=0.6, pad=0.08)
        cbar.ax.tick_params(labelsize=24)
        cbar.set_label('暴雨成灾概率', size=28)
        cbar.set_ticks([0, 0.2, 0.4, 0.6, 0.8, 1])
        cbar.set_ticklabels(['0', '0.2', '0.4', '0.6', '0.8', '1'])

        plt.tight_layout()

        save_path = save_path_template.format(year=year)
        plt.savefig(save_path, dpi=300, bbox_inches='tight')
        print(f"{year}年的图片已保存至：{save_path}")

        plt.close(fig)  # 关闭图形，释放内存

def process_data_and_visualize(excel_path, save_path_template):
    try:
        data = pd.read_excel(excel_path)
    except Exception as e:
        print(f"读取 Excel 文件时出错: {e}")
        return
    features = ['高程', '降水', '土地利用系数', '经度', '纬度']
    data = calculate_disaster_probability(data, features)

    years = [2026, 2030, 2035]
    filtered_data = data[data['年'].isin(years)]

    # 可视化2026, 2030 和2035 年的暴雨成灾概率热力图
    visualize_heatmap(filtered_data, years, save_path_template)


if __name__ == "__main__":
    excel_path = r'E:\predictions\future_predictions_2025_2035_rf1.xlsx'        save_path_template =
r'E:\disaster_probability_heatmap_{year}_dt.png'        process_data_and_visualize(excel_path,
save_path_template)

def visualize_extreme_weather(data, years):
    fig, axes = plt.subplots(3, 1, figsize=(13,30), subplot_kw={'projection': ccrs.PlateCarree()})

    for i, year in enumerate(years):
        ax = axes[i]
```

```
        year_data = data[data['年'] == year]

        ax.set_extent([73, 130, 15, 54], crs=ccrs.PlateCarree())

        ax.add_feature(cfeature.LAND)
        ax.add_feature(cfeature.OCEAN)
        ax.add_feature(cfeature.COASTLINE)
        ax.add_feature(cfeature.LAKES, alpha=0.5)
        ax.add_feature(cfeature.RIVERS)

        extreme_weather_locations = year_data[year_data['is_extreme_weather'] == 1]

        lat_min, lat_max = data['纬度'].min(), data['纬度'].max()
        reversed_latitudes = lat_max + lat_min - extreme_weather_locations['纬度']

        ax.scatter(extreme_weather_locations['经度'], reversed_latitudes,
                   color='red', s=10, alpha=0.7, transform=ccrs.PlateCarree())

        ax.set_title(f"{year}年预测的暴雨灾害地点", fontsize=36)

    plt.tight_layout()
    plt.savefig('extreme_weather_2026_2030_2035_rf.png', dpi=500, bbox_inches='tight')
    plt.show()
```

问题 4

1. 分区

```
import pandas as pd
from shapely.geometry import Point, LineString

def classify_by_qinling_huaihe(longitude, latitude):
    qinling_huaihe_line = LineString([(105.0, 34.0), (120.0, 32.0)])
    location = Point(longitude, latitude)
    return "北" if location.y > qinling_huaihe_line.interpolate(qinling_huaihe_line.project(location)).y
else "南"

def classify_by_huhuan_yong_line(longitude, latitude):
    huhuan_yong_line = LineString([(127.0, 50.0), (98.0, 25.0)])
    location = Point(longitude, latitude)
    return "东" if location.x > huhuan_yong_line.interpolate(huhuan_yong_line.project(location)).x
else "西"

try:
    input_file = 'results_2005_2015.csv'
    print(f"处理文件：{input_file}")

    # 使用GBK 编码读取CSV 文件
    df = pd.read_csv(input_file, encoding='gbk')
    print("成功读取文件")

    df['秦岭淮河线'] = df.apply(lambda row: classify_by_qinling_huaihe(row['经度'], row['纬度']),
axis=1)
    df['胡焕庸线'] = df.apply(lambda row: classify_by_huhuan_yong_line(row['经度'], row['纬度']),
axis=1)
```

```
    print("已添加地理分类列")

    output_file = f"已处理_{input_file}"
    df.to_csv(output_file, index=False, encoding='gbk')

    print(f"处理完成。结果已保存到 '{output_file}'。")

————————————————————可视化分区————————————————————
def plot_china_map_with_regions():
    proj = ccrs.LambertConformal(central_longitude=105, central_latitude=35)
    fig, ax = plt.subplots(figsize=(15, 10), subplot_kw={'projection': proj})
    ax.set_extent([73, 135, 18, 54], crs=ccrs.PlateCarree())

    ax.add_feature(cfeature.LAND)
    ax.add_feature(cfeature.OCEAN)
    ax.add_feature(cfeature.COASTLINE)
    ax.add_feature(cfeature.BORDERS, linestyle=':')

    provinces = cfeature.NaturalEarthFeature(
        category='cultural',
        name='admin_1_states_provinces_lines',
        scale='50m',
        facecolor='none')
    ax.add_feature(provinces, edgecolor='gray')

    qinling_huaihe_line = LineString([(105.0, 34.0), (120.0, 32.0)])
    huhuan_yong_line = LineString([(127.0, 50.0), (98.0, 25.0)])

    ax.plot(*qinling_huaihe_line.xy, transform=ccrs.PlateCarree(), color='black', linewidth=2,
label='秦岭-淮河线')
    ax.plot(*huhuan_yong_line.xy, transform=ccrs.PlateCarree(), color='black', linewidth=2,
label='胡焕庸线')

    intersection_point = qinling_huaihe_line.intersection(huhuan_yong_line)
    if intersection_point:
        ax.plot(intersection_point.x, intersection_point.y, 'ko', transform=ccrs.PlateCarree(),
markersize=10)
        ax.text(intersection_point.x, intersection_point.y, '交叉点', transform=ccrs.PlateCarree(),
            fontsize=12, ha='right', va='bottom')

    # 定义四个区域的多边形
    northeast = Polygon([(120, 32), (135, 32), (135, 54), (127, 50), (120, 32)])
    southeast = Polygon([(120, 32), (135, 32), (135, 18), (98, 25), (105, 34), (120, 32)])
    northwest = Polygon([(73, 54), (127, 50), (105, 34), (73, 34)])
    southwest = Polygon([(73, 34), (105, 34), (98, 25), (73, 18)])

    ax.add_geometries([northeast], ccrs.PlateCarree(), facecolor='lightblue', alpha=0.5, label='东
北')
    ax.add_geometries([southeast], ccrs.PlateCarree(), facecolor='lightgreen', alpha=0.5, label='东
南')
    ax.add_geometries([northwest], ccrs.PlateCarree(), facecolor='lightcoral', alpha=0.5, label='西
北')
    ax.add_geometries([southwest], ccrs.PlateCarree(), facecolor='lightyellow', alpha=0.5,
```

```
label='西南')

    ax.text(125, 45, "东北", transform=ccrs.PlateCarree(), fontsize=14, ha='center', va='center')
    ax.text(125, 25, "东南", transform=ccrs.PlateCarree(), fontsize=14, ha='center', va='center')
    ax.text(90, 40, "西北", transform=ccrs.PlateCarree(), fontsize=14, ha='center', va='center')
    ax.text(90, 25, "西南", transform=ccrs.PlateCarree(), fontsize=14, ha='center', va='center')

    gl = ax.gridlines(crs=ccrs.PlateCarree(), draw_labels=True, linewidth=1, color='gray', alpha=0.5,
linestyle='--')
    gl.top_labels = False
    gl.right_labels = False
    gl.xformatter = LONGITUDE_FORMATTER
    gl.yformatter = LATITUDE_FORMATTER

    ax.legend()
    plt.title('中国地图：四大区域划分', fontsize=16)
    plt.show()

plot_china_map_with_regions()
```

2. 熵权 topsis

```
import pandas as pd
import numpy as np


def load_and_preprocess_data(file_path):
    df = pd.read_csv(file_path, encoding='gbk')
    df['人口'] = df['人口'].replace(0, 1e-6)
    df['GDP'] = df['GDP'].replace(0, np.nan)
    df['log_人口'] = np.log1p(df['人口'])
    df['log_GDP'] = np.log1p(df['GDP'])
    df['人均耕地'] = df['耕地'] * 10000 / df['人口']
    return df


def split_data(df):
    df['区域'] = df.apply(
        lambda row: f"{'东' if row['胡焕庸线'] == '东' else '西'}{'北' if row['秦岭淮河'] == '北' else '南'}",
axis=1)
    return df


def calculate_entropy_weight(data):
    data_normalized = (data - data.min()) / (data.max() - data.min())
    entropy = -np.sum(data_normalized * np.log(data_normalized + 1e-8), axis=0) /
np.log(len(data))
    weights = (1 - entropy) / (len(entropy) - np.sum(entropy))
    return weights / np.sum(weights)


def topsis(data, weights):
    normalized_data = data / np.sqrt((data ** 2).sum(axis=0))
    weighted_normalized = normalized_data * weights
    ideal_best = weighted_normalized.max(axis=0)
    ideal_worst = weighted_normalized.min(axis=0)
```

```
    distance_best = np.sqrt(((weighted_normalized - ideal_best) ** 2).sum(axis=1))
    distance_worst = np.sqrt(((weighted_normalized - ideal_worst) ** 2).sum(axis=1))
    topsis_score = distance_worst / (distance_best + distance_worst)
    return pd.DataFrame({
        'D+': distance_best,
        'D-': distance_worst,
        'C': topsis_score
    })


def analyze_data_by_year_and_region(df):
    columns_to_analyze = ['log_人口', 'log_GDP', '高程', '温度', '人均耕地']

    results = []

    for year in df['年'].unique():
        year_data = df[df['年'] == year]

        weights = calculate_entropy_weight(year_data[columns_to_analyze])
        topsis_results = topsis(year_data[columns_to_analyze], weights)

        topsis_results['年'] = year
        topsis_results['区域'] = year_data['区域'].values

        results.append(topsis_results)

    return pd.concat(results)

if __name__ == "__main__":
    df = load_and_preprocess_data('已处理_results_2005_2015.csv')
    df = split_data(df)

    results = analyze_data_by_year_and_region(df)

    yearly_region_averages = results.groupby(['年', '区域'])[['D+', 'D-', 'C']].mean().reset_index()
    yearly_region_averages = yearly_region_averages.sort_values(['年', '区域'])

    yearly_region_averages['排序结果'] = yearly_region_averages.groupby('年')
['C'].rank(ascending=False).astype(int)

    print("TOPSIS 评价计算结果（年度和区域平均）：")
    print(yearly_region_averages[['年', '区域', 'D+', 'D-', 'C', '排序结果']])

    yearly_region_averages.to_csv('topsis_results_by_year_and_region.csv', index=False,
encoding='utf-8-sig')
    print("\n 分析完成。结果已保存到 'topsis_results_by_year_and_region.csv' 文件中。")

    pivot_results = yearly_region_averages.pivot(index='年', columns='区域', values=['D+', 'D-', 'C',
'排序结果'])
    pivot_results.columns = [f'{col[1]}_{col[0]}' for col in pivot_results.columns]
    pivot_results = pivot_results.reset_index()

    print("\n 重塑后的 TOPSIS 评价计算结果：")
    print(pivot_results)
```

```
    pivot_results.to_csv('topsis_results_pivot.csv', index=False, encoding='utf-8-sig')
    print("\n 重塑后的结果已保存到 'topsis_results_pivot.csv' 文件中。")
```

3. 多元回归

```
import matplotlib.pyplot as plt
from scipy import stats
from io import StringIO

def load_and_preprocess_data(file_path):
    df = pd.read_csv(file_path, encoding='gbk')
    df['人口'] = df['人口'].replace(0, 1e-6)
    df['GDP'] = df['GDP'].replace(0, np.nan)
    df['log_人口'] = np.log1p(df['人口'])
    df['log_GDP'] = np.log1p(df['GDP'])
    df['人均耕地'] = df['耕地']  / df['人口']
    return df

def split_data(df):
    regions = {
        '东北': df[(df['秦岭淮河'] == '北') & (df['胡焕庸线'] == '东')],
        '西北': df[(df['秦岭淮河'] == '北') & (df['胡焕庸线'] == '西')],
        '东南': df[(df['秦岭淮河'] == '南') & (df['胡焕庸线'] == '东')],
        '西南': df[(df['秦岭淮河'] == '南') & (df['胡焕庸线'] == '西')]
    }
    return regions

plt.rcParams['font.sans-serif'] = ['SimHei']
plt.rcParams['axes.unicode_minus'] = False


def perform_regression_analysis(X, y):
    X = sm.add_constant(X)
    model = sm.OLS(y, X).fit()
    return model

def calculate_relative_importance(X, y):
    model = LinearRegression()
    model.fit(X, y)
    return permutation_importance(model, X, y, n_repeats=10, random_state=0)


def analyze_region(region_df, region_name, writer):
    print(f"\n 分析 {region_name} 区域:")

    features = ['高程', '温度', '降水', 'log_人口', 'log_GDP']

    original_stats = region_df[features + ['人均耕地']].describe()
    print(original_stats)
    original_stats.to_excel(writer, sheet_name=f'{region_name}_原始数据统计')

    # 移除包含NaN 的行
    region_df = region_df.dropna(subset=features + ['人均耕地'])
```

```
X = region_df[features]
y = region_df['人均耕地']

scaler = StandardScaler()
X_scaled = pd.DataFrame(scaler.fit_transform(X), columns=X.columns, index=X.index)

# 多元线性回归
model = perform_regression_analysis(X_scaled, y)
print(model.summary())

summary_df = pd.DataFrame({
    'coef': model.params,
    'std err': model.bse,
    't': model.tvalues,
    'P>|t|': model.pvalues,
    '[0.025': model.conf_int()[0],
    '0.975]': model.conf_int()[1]
})
summary_df.to_excel(writer, sheet_name=f'{region_name[:10]}_回归结果')

# 计算部分相关系数
partial_corr = pd.DataFrame(index=features, columns=['Partial Correlation'])
for feature in features:
    others = [f for f in features if f != feature]
    partial_corr.loc[feature, 'Partial Correlation'] = stats.pearsonr(
        model.resid,
        sm.OLS(X_scaled[feature], sm.add_constant(X_scaled[others])).fit().resid
    )[0]

print("\n 部分相关系数:")
print(partial_corr)
partial_corr.to_excel(writer, sheet_name=f'{region_name[:10]}_部分相关系数')

importance = calculate_relative_importance(X_scaled, y)
relative_importance = pd.DataFrame(
    {'Relative Importance': importance.importances_mean},
    index=features
).sort_values('Relative Importance', ascending=False)

print("\n 相对重要性:")
print(relative_importance)
relative_importance.to_excel(writer, sheet_name=f'{region_name[:10]}_相对重要性')

# 分组分析
natural_factors = X_scaled[['高程', '温度', '降水']]
human_factors = X_scaled[['log_人口', 'log_GDP']]

natural_r2 = sm.OLS(y, sm.add_constant(natural_factors)).fit().rsquared
human_r2 = sm.OLS(y, sm.add_constant(human_factors)).fit().rsquared

print(f"\n 自然因素 R-squared: {natural_r2:.4f}")
print(f"人文因素 R-squared: {human_r2:.4f}")

r_squared_df = pd.DataFrame({
```

```
        'Factor': ['自然因素', '人文因素'],
        'R-squared': [natural_r2, human_r2]
    })
    r_squared_df.to_excel(writer, sheet_name=f'{region_name[:10]}_R-squared')


if __name__ == "__main__":
    df = load_and_preprocess_data('已处理_results_2005_2015.csv')
    regions = split_data(df)

    # 创建一个ExcelWriter 对象
    output_folder = "regression_results"
    os.makedirs(output_folder, exist_ok=True)
    excel_path = os.path.join(output_folder, "regression_analysis_results.xlsx")

    with pd.ExcelWriter(excel_path, engine='openpyxl') as writer:
        for region_name, region_df in regions.items():
            try:
                analyze_region(region_df, region_name, writer)
            except Exception as e:
                print(f"处理 {region_name} 时发生错误: {str(e)}")
                continue

    print(f"\n 分析结果已保存到 {excel_path}")
```

4. 基于随机森林分析变量相对重要性程度

```
import pandas as pd
import numpy as np
from scipy import stats
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import r2_score
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt
import seaborn as sns

plt.rcParams['font.sans-serif'] = ['SimHei']
plt.rcParams['axes.unicode_minus'] = False


def load_and_preprocess_data(file_path):
    df = pd.read_csv(file_path, encoding='gbk')
    df['人口'] = df['人口'].replace(0, 1e-6)
    df['GDP'] = df['GDP'].replace(0, np.nan)
    df['log_人口'] = np.log1p(df['人口'])
    df['log_GDP'] = np.log1p(df['GDP'])
    df['人均耕地'] = df['耕地'] / df['人口']
    return df


def split_data(df):
    regions = {
        '东北': df[(df['秦岭淮河'] == '北') & (df['胡焕庸线'] == '东')],
        '西北': df[(df['秦岭淮河'] == '北') & (df['胡焕庸线'] == '西')],
        '东南': df[(df['秦岭淮河'] == '南') & (df['胡焕庸线'] == '东')],
```

```
        '西南': df[(df['秦岭淮河'] == '南') & (df['胡焕庸线'] == '西')]
    }
    return regions


def random_forest_importance(X, y):
    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
    rf = RandomForestRegressor(n_estimators=100, random_state=42)
    rf.fit(X_train, y_train)
    y_pred = rf.predict(X_test)
    r2 = r2_score(y_test, y_pred)
    return rf.feature_importances_, r2


def visualize_importance(importance, columns, region_name):
    plt.figure(figsize=(10, 6))
    sns.barplot(x=importance, y=columns, palette='viridis')
    plt.title(f'Feature Importance in {region_name} Region')
    plt.xlabel('Importance')
    plt.ylabel('Features')
    plt.tight_layout()
    plt.show()


def visualize_spearman_correlation(df, columns, region_name):
    corr = df[columns].corr(method='spearman')
    plt.figure(figsize=(10, 8))
    sns.heatmap(corr, annot=True, cmap='coolwarm', vmin=-1, vmax=1, center=0)
    plt.title(f'Spearman Correlation Matrix in {region_name} Region')
    plt.tight_layout()
    plt.show()


def analyze_region(region_df, region_name):
    print(f"\n 分析 {region_name} 区域:")

    columns_to_analyze = ['log_人口', 'log_GDP', '高程', '温度', '人均耕地']
    correlations = region_df[columns_to_analyze].corr(method='spearman')['人均耕地'].drop('人均耕
地')
    print("与人均耕地的 Spearman 相关系数:")
    for col, corr in correlations.items():
        print(f"{col}: {corr:.4f}")

    # 随机森林分析
    X = region_df[columns_to_analyze[:-1]]  # 不包括 '人均耕地'
    y = region_df['人均耕地']
    importance, r2 = random_forest_importance(X, y)

    print("\n 随机森林特征重要性:")
    for col, imp in zip(columns_to_analyze[:-1], importance):
        print(f"{col}: {imp:.4f}")

print(f"\n 随机森林模型 R² 评分: {r2:.4f}")

    visualize_importance(importance, columns_to_analyze[:-1], region_name)
```

```
    visualize_spearman_correlation(region_df, columns_to_analyze, region_name)

if __name__ == "__main__":
    df = load_and_preprocess_data('已处理_results_2005_2015.csv')
    regions = split_data(df)

    for region_name, region_df in regions.items():
        analyze_region(region_df, region_name)
```